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ACADEMIC NETWORK FOR A GREEN & INNOVATIVE EUROPE

WP2–Setting-up the Graduate Study Program (GSP) "Transition to Green for Sustainable Society"

Academic Network for a Green and Innovative Europe

D2.1.1: Report on inventory and analysis of required topics

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1 Foreword

Like any new study program, the "Transition to Green for Sustainable Society" program should start with an analysis based on the needs/deficiencies identified in society. In the present case, the need is triple:

1. The current messages related to climate change do not have the expected impact in the society.
2. The current messages are often sent uncoordinatedly.
3. The green transition of the WATERIC fields is hampered by the lack of understanding, trust and interest of the specialists in these fields regarding the measures that must be taken to initiate and accelerate the transition towards an ecological manufacturing and services.

Some of these problems where already identified when and establishing the structure of the ANGIE network, being at this stage necessary to identify those skills that are useful to WATERIC specialists, trainers and stakeholders, who have decision-making power regarding the transition to green by lowering-carbon footprint.

In summary this activity aims at conducting an Inventory of the most relevant climate problem identified in WATERIC fields and an overview of the actual situation of the partners countries in terms of “Transition to Green for Sustainable Society” to understand, analyse and identify the most relevant topics to be used within the curriculum [1].

2 Overview

The need to foster a healthy coexistence between human activities and the planet is evident in an era where environmental sustainability imperatives are more pressing than ever. The "Transition to Green for Sustainable Society" programme is a shining example of change as it acknowledges the urgent need for a paradigm shift towards a more sustainable society. This innovative study programme is a reaction to a critical need that has been discovered in our global community, not just an academic endeavour.

Society finds themselves at a crossroads when it comes to environmental issues, such as resource depletion and climate change, and there is a major gap in our understanding and implementation of sustainable practices. The "Transition to Green" programme is an intentional attempt to close this gap by addressing the need for comprehensive education and workable solutions that will help students, teachers, stakeholder to move towards a more sustainable and greener future.

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This deliverable intends to analyse the genesis of this innovative program, exploring its roots in the identified deficiencies within contemporary society. By understanding the foundations of the "Transition to Green for Sustainable Society" initiative, Angie project aims to contribute to a sustainable and resilient future.

In order to reach this goal spreadsheets, figure 1, were prepared and distributed among each partner to identify the problems needs of the 8 partners countries, namely Romania, Austria, Spain, Italy, Slovenia, Slovakia, Germany and Portugal, in the different 7 sectors identified, Agriculture, waste, transport, energy production, residential activities and industrial activities.

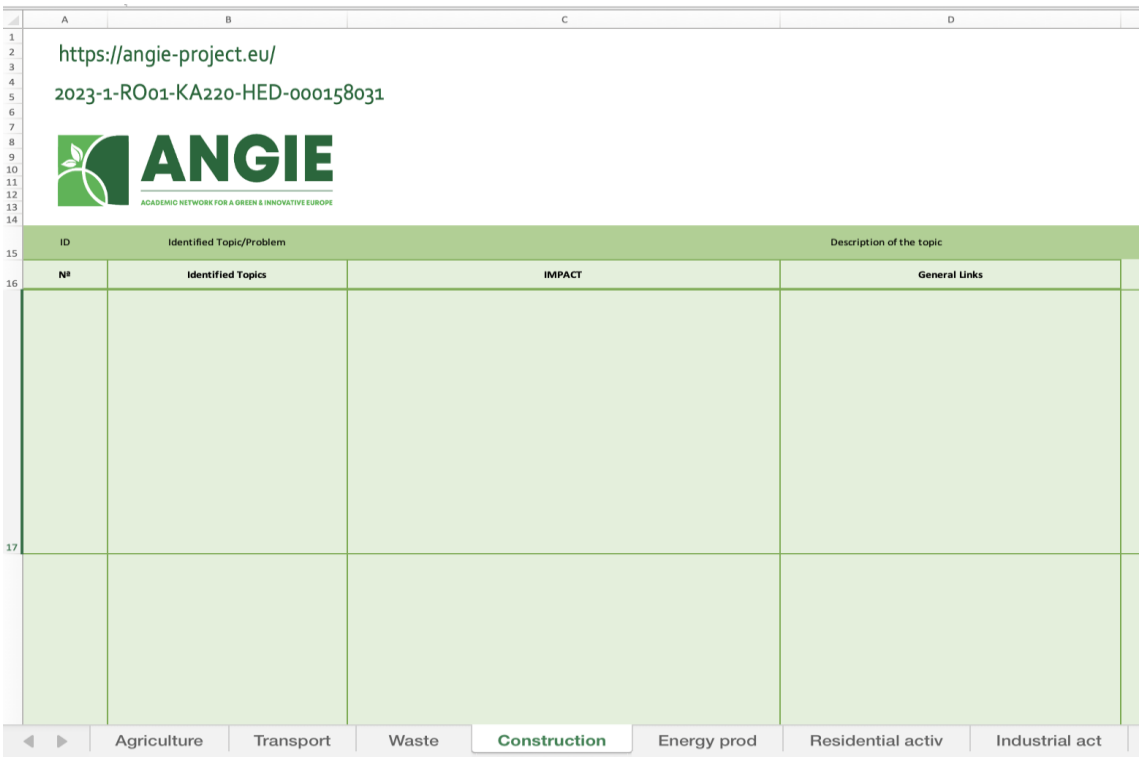

Figure 1. Spreadsheets distributed among each partner to identify the Climate problems in the WATERIC field in the different countries.

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3 Problems and Needs Identified by each partner in the 7 sectors.

Based on the analysis of the spreadsheets provided by each partner, on the WATERIC field, namely: agriculture, waste, transport, energy production, residential activities, and industrial activities a summary of the problems and needs was carried out. The results obtained ate aligned with the 2030 Climate targets by the EU (fig 2).

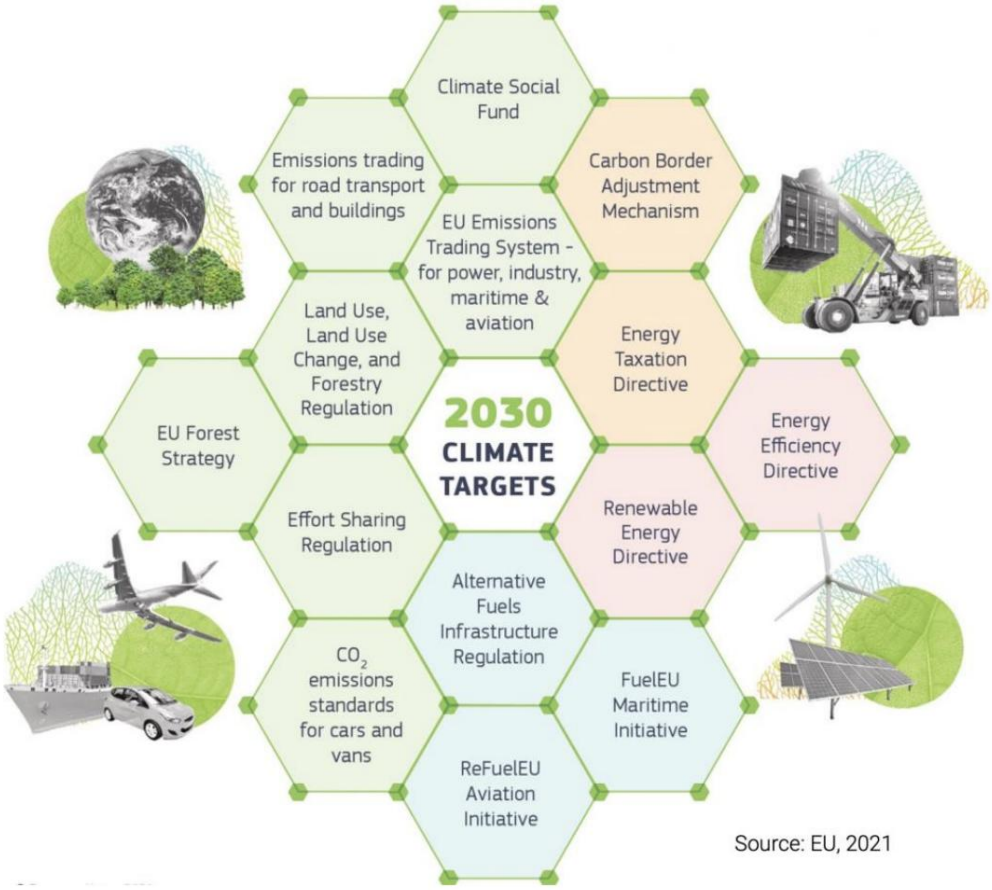

Figure2 - Schematic overview illustrating the different elements of the Fit for 55 packages. [2].SOURCE EU, 2021(file:///Users/inespires/Documents/projectos/ANGIE/PR2/graficos/ETC%20Report%20PaMs.pdf)

3.1 Agriculture

Table 1 indicates a summary of the problems and needs identified by the ANGIE partners in the scope of agriculture reaching “climate targets”.

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Table 1 - summary of the problems and needs identified by the ANGIE partners in the scope of agriculture.

Identified Topics	IMPACT
Pesticide Use: Pesticides can have negative effects on ecosystems, including the decline of beneficial insects and contamination of water sources. Minimizing the use of harmful pesticides and promoting organic farming for healthier ecosystems/Organic farming and integrated pest management aim to reduce reliance on harmful pesticides.	Worldwide it is estimated that approximately 1.8 billion people engage in agriculture and most use pesticides to protect the food and commercial products that they produce. Pesticides can contaminate soil and water, leading to adverse effects on non-target organisms and ecosystems. Transitioning to integrated pest management and organic farming can reduce pesticide-related environmental harm. Source: https://www.ewg.org/search?fullsearch=pesticides+in+use
Water consumption/withdrawals in agriculture: Agriculture is a major consumer of freshwater resources. Inefficient irrigation practices and water-intensive crops can contribute to water scarcity.	Currently 70% of global groundwater withdrawals, and even more in arid and semi-arid regions, are used in the agricultural production of food, fibres, livestock and industrial crops, and an estimated 38% of the lands equipped for irrigation is serviced by this resource. In order to meet global water and agricultural demands by 2050, including an estimated 50% increase in food, the development of groundwater could act as a catalyst for improving agricultural productivity and economic growth by increasing the extent of irrigated areas. Source: https://www.unwater.org/water-facts
Agriculture Sustainability: Excessive use of synthetic fertilizers and pesticides can lead to soil degradation, water pollution, and harm to biodiversity. The need to transition towards sustainable and environmentally friendly farming practices to ensure long-term food security/Sustainable agriculture promotes practices that maintain or enhance soil health and minimize environmental impact. Unsustainable agricultural practices contribute to soil erosion, loss of biodiversity, and water pollution. Implementing sustainable agriculture can lead to healthier	The Global Sustainable Agriculture Market size is valued at 13.32 billion in 2022 and is predicted to reach 31.35 billion by the year 2031 at a 10.17% CAGR during the forecast period for 2023-2031. Sustaining agricultural production depends on the quality and availability of natural resources like soil and water. Source: https://www.worldwildlife.org/industries/sustainable-agriculture https://www.insightaceanalytic.com/report/sustainable-agriculture-market/1661

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ecosystems, improved soil quality, and reduced environmental impact	

Soil and water are two examples of natural resources whose availability and quality determine the productivity of sustainable agriculture. By promoting the conservation and sustainable use of these priceless natural resources through appropriate and area-specific strategies, agriculture expansion may be encouraged. Systems of agriculture that are sustainable are those that are profitable, satisfy the need of the public for wholesome food, and protect or improve the environment's resources for future generations.

It is well known that climate change and the weather have an immediate effect on agricultural productivity. Under normal circumstances, crop development, growth, water use, and yield are all influenced by the climate during the growing season. Even a small deviation from the norm might have a detrimental impact on crop output. The hydrological cycle has been impacted by global warming, and this influences the amount and frequency of rainfall. Water availability becomes the most crucial factor in agriculture since the majority of fertile land lacks irrigation infrastructure.

Because of how climate change affects agriculture, forest protection will also influence agriculture's sustainability. The single policies and measures (PaMs) with regard to the environmental impact are displayed in figure 3. The amounts of greenhouse gases in the atmosphere must quickly stabilise in order to prevent catastrophic climate impacts. Reducing greenhouse gas emissions globally will be necessary, but so will policies and technology aimed at removing carbon from the atmosphere. Increased carbon removal from carbon removal technologies as well as from the Land Use, Land-Use Change, and Forestry (LULUCF) sector will be necessary to achieve this removal.

Naturally, plants take in CO_2 from the atmosphere and utilise it to develop. Because of this, planting new trees and afforestation will boost the removal of carbon dioxide from the atmosphere. In addition to plants, soils are also carbon sinks. By altering farming methods, soils can absorb more carbon, so absorbing carbon out of the atmosphere.

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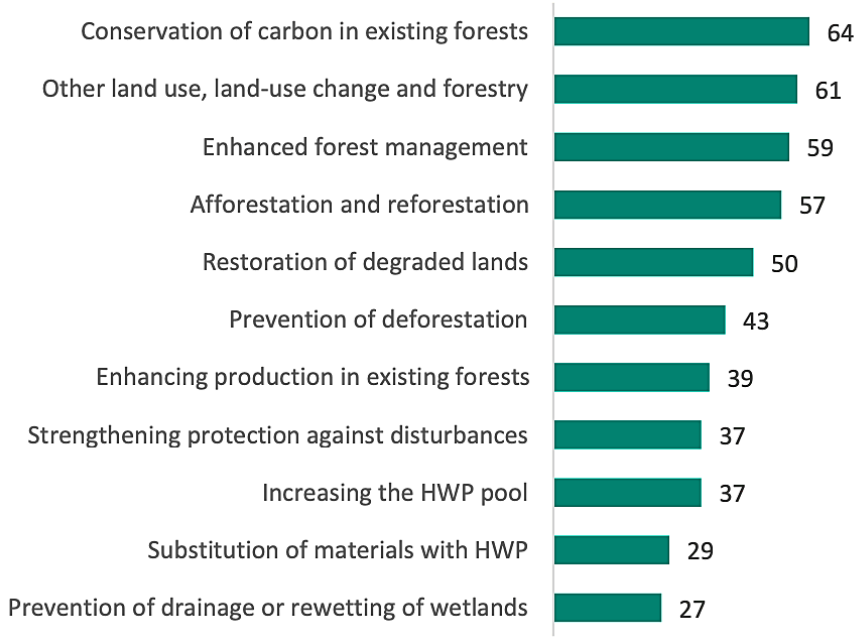

Figure 3. Number of single policies and measures (PaMs) per objective in the LULUCF sector. source:" Overview of reported integrated national climate and energy policies and measures in Europe in 2023" ETC-CM Report 2023/06 [4].

3.2 Transport

Table 2 indicates a summary of the problems and needs identified by the ANGIE partners in the scope transport to reach "climate targets".

Table 2 - summary of the problems and needs identified by the ANGIE partners in the scope transport.

Identified Topics	IMPACT
Emissions of fuels combustion/greenhouse gas emissions. Combustion of fossil fuels in transportation contributes to air pollution and climate change. Electric vehicles, improved fuel efficiency, and public transportation can help mitigate these impacts.	Global CO_2 emissions from transport by sub-sector in the Net Zero Scenario, 2000-2030. In 2022 global CO_2 emissions from the transport sector grew by more than 250 Mt CO2 to nearly 8 Gt CO_2, 3% more than in 2021.The transportation sector is a major contributor to air pollution and climate change. Greenhouse gas emissions from vehicles have a direct impact on air quality and global temperatures Source: https://www.epa.gov/transportation-air-pollution-and-climate-change

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Inadequate transportation infrastructure can lead to traffic congestion and increased fuel consumption. Investing in efficient public transportation and smart infrastructure planning can address these issues.	Transport is at the center of many economic and social development challenges, accounting for about 64% of global oil consumption, 27% of all energy use, and 23% of the world's energy-related carbon dioxide emissions. When transport systems are deficient in terms of capacity or reliability, they can have an economic cost, such as reduced or missed opportunities and lower quality of life. At the aggregate level, efficient transportation reduces costs in many economic sectors, while inefficient transportation increases these costs. Inadequate transportation infrastructure contributes to traffic congestion, resulting in increased fuel consumption, time wastage, and air pollution Source: https://www.wri.org/videos/8-questions-sustainable-urban-development-luis-bettencourt https://energypedia.info/wiki/The_Environment_and_Safety:_Adverse_Effects_of_Transport_Infrastructure
Alternative Fuels/sustainable transportation. Transitioning to alternative fuels, such as electricity, hydrogen, or biofuels, can reduce the carbon footprint of transportation. Promoting the use of electric vehicles and other alternative fuels to reduce dependence on fossil fuels. There are four major categories of transportation energy sources The major transportation energy sources used in the United States are: Petroleum products made from crude oil and from natural gas processing, including motor and aviation gasoline, distillate fuels (mostly diesel fuel), jet fuel, residual fuel oil, and propane; Biofuels; Natural gas; Electricity produced from many different energy sources	Transportation emits billions of tonnes of greenhouse gases into the atmosphere every year. These gases primarily come from the burning of petroleum-based fuels, such as diesel and gasoline, in internal combustion engine vehicles (ICEVs). Since 1970, transportation emissions have almost tripled due to growing populations and economies. The sector now accounts for 20 percent of global CO_2 emissions and is the second-largest carbon-polluting sector worldwide. Although the outbreak of COVID-19 caused global transportation emissions to plummet in 2020, sector emissions rebounded to 7.6 Gt CO_2 in 2021 and are expected to have grown a further three percent in 2022. Source: https://www.wri.org/cities/urban-mobility

3.3 Waste

Table 3 indicates a summary of the problems and needs identified by the ANGIE partners in the scope of waste to reach "climate targets".

Table 3 - summary of the problems and needs identified by the ANGIE partners in the scope of waste.

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Identified Topics	IMPACT
Improper waste disposal and landfill use contribute to environmental pollution. Effective waste management involves recycling, composting, and reducing overall waste generation.	**World Waste Index 2022**: 417 kg of waste per capita ends up in landfill and is disposed of in a particularly environmentally harmful way. The world generates 2.01 billion tonnes of municipal solid waste annually, this is expected to increase by roughly 70 percent to 3.4 billion metric tonnes by 2050. Source: https://www.epa.gov/recycle
Plastic Pollution: Improper disposal of plastic waste leads to ocean pollution and harm to marine life. Initiatives to reduce single-use plastics and promote recycling are essential. Implementing effective waste management systems to reduce landfill usage and promote recycling, effective waste management leads to pollution, habitat destruction, and health hazards. Proper waste management practices, including recycling and waste reduction, can alleviate these issues.	from the OECD's Global Plastic Outlook (2022). Nearly one-quarter of the world's plastic waste is mismanaged or littered. Around 82 million tonnes. This means it's not stored in secure landfills, recycled or incinerated. Millions of tonnes of plastic enter oceans annually, harming marine life and ecosystems. Efforts to reduce single-use plastics and promote recycling are critical for addressing this global problem. Source: https://www.plasticpollutioncoalition.org/
Circular Economy; recycling, reuse and reduce. Shifting towards a circular economy model aims to minimize waste by reusing, repairing, and recycling materials. Encouraging a shift towards a circular economy to minimize waste and maximize resource efficiency.	The global economy is now only 7.2% circular The global situation is getting worse year on year—driven by rising material extraction and use. Rising material extraction has shrunk global circularity: from 9.1% in 2018, to 8.6% 2020, and now 7.2% in 2023. Embracing a circular economy model can significantly reduce the environmental impact of waste by promoting reuse and recycling. Materials that are not recycled are lost at the end of the economic cycle, as are the embedded energy and other resources used in their processing. Embedded energy refers to the energy that was needed to create the product. Recycling is the treatment method by which waste materials are actually returned into the economic cycle and

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	therefore, can continue creating value. This area comprises 2 statistical indicators: overall recycling rates recycling rates for specific waste streams. Increasing the amount of waste that is recycled is part of the transition to a circular economy. The overall recycling rate helps to monitor progress in this area. In addition, the recycling rates for specific waste streams provide more detailed insights, in particular about packaging waste, including plastic, and electrical and electronic waste, which pose a challenge to the economy and the environment. Source: https://www.ellenmacarthurfoundation.org/topics/circular-economy-introduction/overview

3.4 Construction

Table 4 indicates a summary of the problems and needs identified by the ANGIE partners in the scope of construction to reach "climate targets".

Table 4 - summary of the problems and needs identified by the ANGIE partners in the scope of construction

Identified Topics	IMPACT
Energy Efficiency: Inefficient building designs and materials contribute to high energy consumption. Implementing energy-efficient technologies and designs can reduce the environmental impact of construction.	Buildings account for a substantial portion of global energy consumption. Improving energy efficiency in construction reduces greenhouse gas emissions and lowers energy costs. The operations of buildings account for 30% of global final energy consumption and 26% of global energy-related emissions1 (8% being direct emissions in buildings and 18% indirect emissions from the production of electricity and heat used in buildings). Direct emissions from the buildings sector decreased in 2022 compared to the year before, despite extreme temperatures driving up heating-related emissions in certain regions. In 2022, buildings sector energy use increased by around 1%. Source: https://www.wri.org/cities/urban-efficiency-climate

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Sustainable Materials: onventional construction materials, such as cement and steel, have a significant environmental footprint. Sustainable alternatives, like bamboo or recycled materials, can be more environmentally friendly.	Traditional construction materials contribute to deforestation, energy consumption, and pollution. Sustainable materials reduce the environmental footprint of construction projects. Source: https://www.energy.gov/eere/ammto/advanced-materials-manufacturing-technologies-office https://www.usgbc.org/
Sustainable urban planning: Poorly planned urban development can lead to habitat destruction and increased energy consumption. Sustainable urban planning focuses on creating livable, efficient, and green cities.	Top 5 Most Sustainable Cities. Oslo, Norway. Stockholm, Sweden. Copenhagen, Denmark. Lahti, Finland. London, UK. https://www.energy.gov/save https://unhabitat.org/ https://www.energy.gov/eere/energy-efficiency-buildings-and-industry

3.5 Energy Production

Table 5 indicates a summary of the problems and needs identified by the ANGIE partners in the scope of Energy production to reach "climate targets".

Table 5 - summary of the problems and needs identified by the ANGIE partners in the scope of Energy production.

Identified Topics	IMPACT
Renewable Energy: Fossil fuel-based energy production is a major contributor to climate change. Transitioning to renewable energy sources like solar, wind, and hydro reduces greenhouse gas emissions. Increasing the share of renewable energy sources (solar, wind, hydro) to reduce reliance on fossil fuels. Becoming the world's first climate-neutral continent by 2050 is the objective behind the European Green Deal (COM(2019) 640 final), the very ambitious package of measures that should enable European citizens and businesses to benefit from sustainable green transition. The use of renewable energy has many	In 2021, renewable energy represented 21.8 % of energy consumed in the EU, down from 22.1% in 2020. Transitioning to renewable energy sources reduces dependence on fossil fuels, mitigating climate change and air pollution. Source: https://www.iea.org/topics/renewable-integration

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potential benefits, including a reduction in greenhouse gas emissions, the diversification of energy supplies and a reduced dependency on fossil fuel markets (in particular, oil and gas). The growth of renewable energy sources may also stimulate employment in the EU, through the creation of jobs in new 'green' technologies.	
Energy Storage: The intermittent nature of renewable energy sources requires effective energy storage solutions. Advancements in battery technology and other storage methods are crucial. Developing efficient energy storage solutions to address the intermittent nature of renewable energy. Efficient energy storage solutions are crucial for optimizing the use of intermittent renewable energy sources and enhancing grid stability. Depending on technology, the market is categorized into pumped hydro storage, battery energy storage, compressed air energy storage, and flywheel energy storage.	Global electricity demand is set to more than double by mid-century, relative to 2020 levels. With renewable sources – particularly wind and solar – expected to account for the largest share of power output in the coming decades, energy storage will play a significant role in maintaining the balance between supply and demand. Global investments in energy storage and power grids surpassed 337 billion U.S. dollars in 2022 and the market is forecast to continue growing. Pumped hydro, hydrogen, batteries, and thermal storage are a few of the technologies currently in the spotlight. Source: https://www.energy.gov/energy-storage
Grid Modernization: Upgrading power grids for improved efficiency and reliability. For example, using smart grids: Smart grids are electricity network that use digital technologies, sensors and software to better match the supply and demand of electricity in real time while minimizing costs and maintaining the stability and reliability of the grid.	The smart grid technology market was estimated at almost 50 billion U.S. dollars in 2022 and is forecast to grow at a Compound-Annual Growth Rate (CAGR) of 17.4 percent until 2028, to reach roughly 130 billion U.S. dollars. The term smart grid covers a wide range of technologies and areas, including Advanced Metering Infrastructure (AMI), network and distribution management, and electricity supply security. Source: https://www.energy.gov/oe/office-electricity; https://www.iea.org/energy-system/electricity/smart-grids

1.1 Residential activities

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Table 6 indicates a summary of the problems and needs identified by the ANGIE partners in the scope of residential activities to reach “climate targets”.

Table 6 - summary of the problems and needs identified by the ANGIE partners in the scope of residential activities.

Identified Topics	IMPACT
Energy Consumption: Inefficient appliances and heating/cooling systems contribute to high energy consumption. Implementing energy-efficient practices and technologies can reduce household energy use.	In 2021, households represented 27% of final energy consumption, or 18.6% of gross inland energy consumption, in the EU. Energy-efficient practices in households reduce electricity bills and lower overall energy demand, contributing to environmental sustainability. n 2021, natural gas accounted for 33.5% of the EU final energy consumption in households, electricity for 24.6%, renewables and wastes for 21.2% and oil&petroleum products for 9.5%. Derived heat for 8.6% and a small proportion, 2.5%, was still covered by coal products (solid fossil fuels). Source: https://www.energy.gov/energysaver/energy-saver

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Waste production/Reduction: Excessive use of single-use products and improper waste disposal contribute to pollution. Adopting recycling and reducing consumption of disposable items can mitigate these issues. Household waste contributes to landfills and pollution. Reducing waste through recycling and conscious consumption minimizes environmental impact.	waste generation has increased massively around the world in recent decades, and there are no signs of it slowing down. By 2050, worldwide municipal solid waste generation is expected to have increased by roughly 70 percent to 3.4 billion metric tonnes. This is due to a number of factors, such as population growth, urbanization, and economic growth, as well as consumer shopping habits. With such immense volumes of waste arising, the need for authorities to provide adequate waste treatment and disposal services has become ever more important. However, less than 20 percent of waste is recycled each year, with huge quantities still sent to landfill sites. Waste is also often disposed of at hazardous open dump sites, especially in developing nations. Richer countries produce more waste than poorer countries, but typically have better waste management to help deal with these issues. Show more promoting recycling and reducing single-use products in households. Source: https://www.epa.gov/recycle
Water waste/Conservation: Wasteful water practices in households contribute to water scarcity. Water-saving technologies and awareness campaigns can promote responsible water use	Worldwide, we use more than 4 trillion cubic meters of freshwater per year. Among the EU countries, Croatia recorded the highest renewable freshwater resources (with a long-term average of 30 000 m^3 per inhabitant), followed by Finland (19 900 m^3), Latvia (19 400 m^3) and Sweden (18 900 m^3). Responsible water use in homes helps conserve water resources, particularly in regions facing water scarcity. Source: https://www.epa.gov/watersense

3.6 Industrial activities

Table 7 indicates a summary of the problems and needs identified by the ANGIE partners in the scope of Industrial activities to reach “climate targets”.

Table 7 - summary of the problems and needs identified by the ANGIE partners in the scope of Industrial activities.

Identified Topics	IMPACT

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Greenhouse gases Emissions: Industrial processes release significant amounts of pollutants and greenhouse gases. Implementing cleaner production methods and adopting technologies for emissions reduction is critical. reenhouse gas (GHG) emissions encompass more than just carbon dioxide (CO_2), even though CO_2 is the most prevalent and widely recognized type. In fact, other greenhouse gases such as methane (CH4) and nitrous oxide (N2O) are even more potent in terms of their impact on global warming. These gasses, although often found in smaller quantities, can have a more substantial warming effect on the Earth's atmosphere, sometimes many times greater than CO_2.	The industry sector in 2022 was directly responsible for emitting 9.0 Gt of CO_2, accounting for a quarter of global energy system CO_2 emissions. Annual emissions slightly declined in both 2020 and 2022, but not enough to align with the Net Zero Emissions by 2050 (NZE) Scenario, in which industrial emissions fall to about 7 Gt CO_2 by 2030. Industrial emissions contribute to air pollution and climate change. Implementing emission reduction technologies and cleaner production methods is essential. Source: https://www.epa.gov/p2
Resource Efficiency/circulat ecomony: Inefficient use of resources in industrial production contributes to environmental degradation. Adopting circular economy principles and sustainable manufacturing practices is essential.	Maximizing resource efficiency in industrial processes minimizes environmental degradation and reduces the demand for raw materials. The worldwide revenue of circular economy transactions, comprised of the categories secondhand, rental, and refurbished goods, was estimated to total roughly 339 billion U.S. dollars in 2022. This was forecast to more than double by 2026. Source: https://www.energy.gov/eere/energy-efficiency-buildings-and-industry https://www.unido.org/
Environmental Performance Index (EPI) and Environmental Compliance: Failure to comply with environmental regulations can result in pollution and habitat destruction. Strict enforcement and adherence to environmental standards are crucial. Environmental Performance Index (EPI) provides a data-driven summary of the state of sustainability around the world. Using 40 performance indicators across 11 issue categories, the EPI ranks 180 countries on climate change performance, environmental health, and ecosystem vitality. These indicators provide a gauge at a national scale of how close countries are to established environmental policy targets. The EPI offers a scorecard that highlights leaders and laggards in environmental performance and provides practical guidance for countries that aspire to move toward a sustainable future	Based on the Environmental Performance Index (EPI) values, countries in the global west were the world's greenest in 2022. That year, Denmark had the highest level of environmental performance worldwide with an EPI value of 77.9. Source: https://www.epa.gov/enforcement https://www3.weforum.org/docs/WEF_The_Global_Risks_Report_2022.pdf

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3.7 Comments on the Climate change needs.

Each Member State should plan how to achieve their respective national objectives and contributions to EU targets across the five dimensions of the Energy Union by 2030. These five dimensions are [4].

- Decarbonisation,
- Energy efficiency,
- Energy security,
- Internal energy market, and
- Research, innovation and competitiveness.

Most of the policies in place are related to decarbonisation flowed by energy efficiency (figure 3)

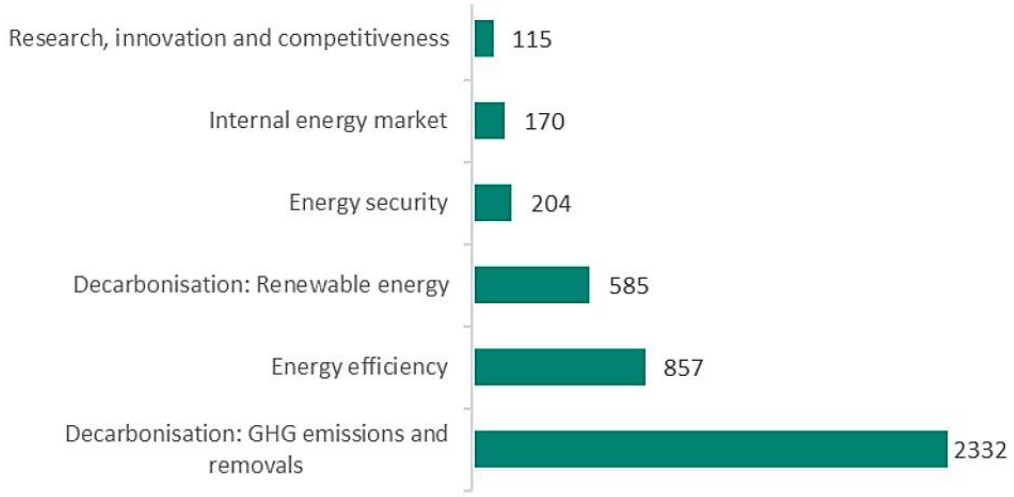

Figure 3- Number of policies and measures per dimension reported by Member States [4].

The policies and measures associated with research, innovation and competitiveness are distributed between the R&I priorities represented in Figure 4. The most common priorities are nº1 in renewable energy and energy efficiency.

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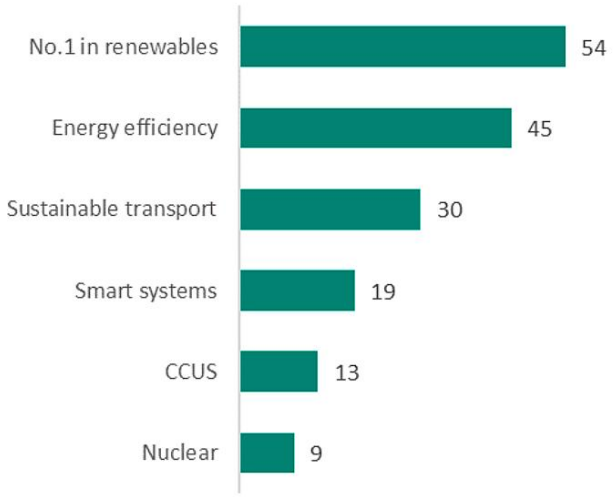

Figure 4 - Distribution of Priority and measures associated with the Research, innovation and competitiveness dimension between the Energy Union R&I Priorities [4].

To prevent the worst impacts of climate change, European countries have developed national strategies, policies and accompanying measures to mitigate climate change. These include for example targets for greenhouse gas emissions in key sectors of the economy, promoting the use of renewable energy and low carbon fuels or improvements in the energy efficiency of buildings (Figure 5).

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No.1 in renewables	Smart systems	Energy efficiency	Sustainable transport	CCUS	Nuclear
Photovoltaics	Advanced metering infrastructure (including smart meters)	Integration of renewables in buildings	Batteries	CCS	Nuclear fusion
Geothermal heat & power	Building/home energy management systems	Thermal performance of building	Electric vehicles	CCU	Small modular reactors
Wind (onshore and offshore)	Grid-scale energy storage	Appliances & lighting	Road transport (non-EV)	Other – CCUS	Decommissioning of nuclear installations
Ocean energy	Direct current (DC) technologies	Heat pumps	Rail transport		Nuclear fission (focus on safety, security, radiation protection)
Concentrated solar power & heat, solar thermal electricity	EV charging infrastructure	Other HVAC (not heat pumps)	Shipping		Other - Nuclear
Hydropower and pumped hydro-storage	Efficient (smart) electrical power transmission	Energy efficiency in industry	Aviation		
Renewable fuels of non-biological origin (other than hydrogen)	Efficient (smart) electrical power distribution	Industrial and district heat & cold management	(Sustainable transport) Advanced biofuels		
Bioenergy	Other - Smart systems	Other - Energy efficiency	Solar fuels (direct)		
Sustainable biofuels			Renewable hydrogen		
Advanced biofuels			Renewable fuels of non-biological origin (other)		
Direct Solar fuels			Other - Sustainable transport		
Other - No 1 in renewables					

Source: ETC CM, 2023

Figure 5- Supported clean energy/low carbon technology options associated with each Energy Union R&I Priority[4].

Universal access to affordable, reliable and sustainable energy should be achieved. This includes improving energy efficiency, increasing the share of renewables and further diversifying the energy mix while ensuring affordability of energy for citizens. Monitoring of this policies involves looking at developments in energy consumption, energy supply and access to affordable energy. Additionally, efforts should be made in order to reach the following:

- Thermal Improvement of Building Stock
- Climate Neutral New Buildings
- Modal shift in passenger and freight transport to environmentally friendly transport modes

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- Increase fuel efficiency in road transport
- Increase the share of clean energy sources in road transport
- Increase energy efficiency in energy and manufacturing industries
- Increase the share of renewable energy in power supply and district heating

4 Statistics related to green transition for each partner country

Most of the policies and measures put in place by each country affect the “the way” and magnitude on how the climate targets are obtained.

A summary of some results available per partner´s country (Romania, Spain, Italy Portugal, Slovenia, Austria, Slovakia and Germany) will be presented in the following paragraphs, by WATERIC domain.

4.1 Agriculture

As referred previously EU policies and measures seeks to protect, restore and promote the conservation and sustainable use of terrestrial ecosystems. This includes efforts to sustainably manage forests and halt deforestation, combat desertification, restore degraded land and soil, halt biodiversity loss and protect threatened species.
Statistics, from the 8 ANGIE partner´s countries, on *Sales of pesticides*, *share of irrigable and irrigated areas in utilised agricultural area (UAA)*, *Final energy consumption by agriculture/forestry per hectare of utilised agricultural area*, *Ammonia emissions from agriculture and Greenhouse gas emissions from agriculture*, will be presented in the next subsections.

4.1.1 Sales of pesticides

The volume of pesticides sold in the ANGIE Partners countries, can be seen in figure 6. This data collection comprises pesticide sales data based on the term "placing on the market, plant protection products ". From that chart it can be seen that Spain in the country that consumes more pesticides, while Slovenia the lowest.

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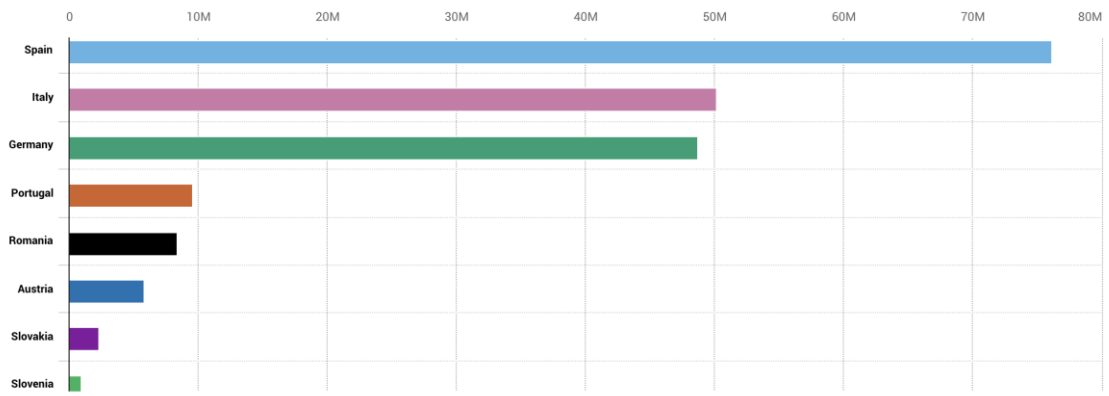

Figure 6 - Volume of pesticides sold in the 8 partners countries in 2021. Units are in Kg [5].

4.1.2 Share of irrigable and irrigated areas in utilised agricultural area (UAA)

The primary objective of irrigation is to provide plants with sufficient water to obtain optimum yields and a high-quality harvested crop. Irrigation is essential in dry climates where precipitation is not sufficient for plant growth. Supplemental irrigation is also vital to produce high levels of quality and yield of crops on croplands in semi-arid and sub-humid climates during dry seasonal periods and drought spells where otherwise farming would not be economically viable. Increasingly dry summers due to climate change sharpen the situation. Irrigation increases yield and therefore contributes significantly to agricultural output and food supply.

At the same time, irrigation water use is a major driving force behind water abstraction. The need for a more sustainable and integrated approach to managing water resources in Europe is reflected in water-related policy and legislation. The Water Framework Directive (WFD) (Directive 2000/60/EC) requires the 'promotion of sustainable water use based on long term protection of available water resources'. In recognition of the acuteness of the water scarcity and drought challenges in Europe, the European Commission undertook in 2006 and early 2007 an in-depth assessment of the situation at EU level. In July 2007 the Commission adopted a Communication on Water Scarcity and Droughts, which identified an initial set of policy options to be taken at European, national and regional levels to address water scarcity within the EU.

Also the Common Agricultural Policy (CAP) instruments aim at a sustainable water management. The indicator on irrigation is part of the Agri-environmental indicators (AEIs) of the European Union (EU) which track the integration of environmental concerns into the CAP at EU, national and regional levels [5].

Figure 7 shows the share of irrigable and irrigated area in utilised agricultural area (UAA) in the 8 partners' countries. The irrigable area is the area which is equipped for irrigation and gives a long-term view of the irrigation needs and expected irrigation needs of farmers. The irrigated area measures the actual amount of land irrigated and is a snapshot of the irrigation needed in that particular year. Crops under glass and kitchen

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gardens, which are assumed to be generally irrigable and irrigated, are not considered within the data collection.

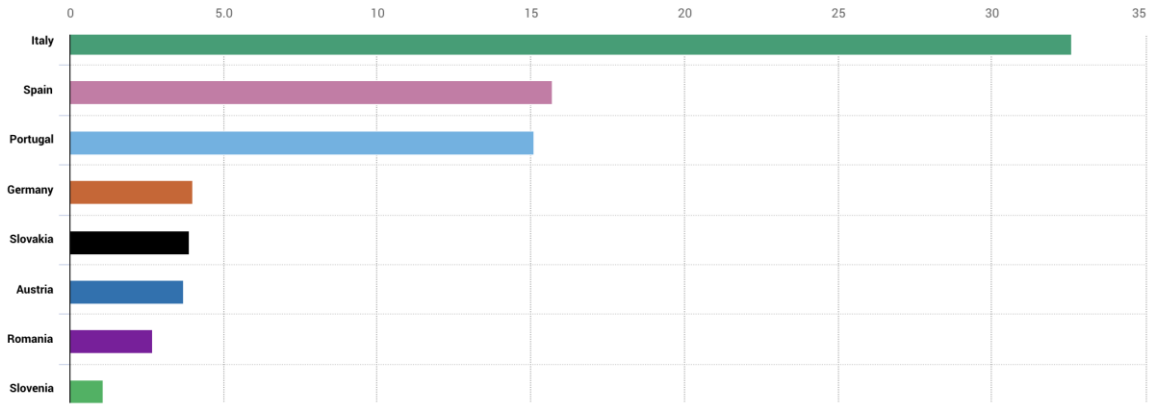

figure 7 - Share of irrigable and irrigated areas in the utilised agricultural area (UAA); the irrigable area is the area, which is equipped for irrigation, while the irrigated area measures the actual amount of land irrigated; based on Farm Structure Survey data, 2016, Units of measure: Percentage of utilized agricultural area (UAA) [5].

4.1.3 Final energy consumption by agriculture/forestry per hectare of utilised agricultural area

In order to produce agricultural products, energy from fossil fuels is necessary. Different crop production techniques need the direct consumption of fuel and electricity, as well as the indirect usage of energy needed to produce agricultural inputs like pesticides and fertilisers. Although direct energy use is frequently discussed when discussing energy usage in agriculture, it is important to note that indirect energy uses, such as the manufacturing of nitrogen fertiliser, account for 50% or more of overall energy consumption.

In fact, a variety of indirect energy inputs, such as synthetic inorganic fertilisers and insecticides, along with seeds, feed, and specialty feed supplements for livestock, influence the energy efficiency of the feed and food production process [6].

Figure 8 represents the final energy consumption by agriculture/forestry per hectare of utilised agricultural area in the 8 partners' countries. This indicator expresses the sum of the energy supplied to agriculture for all energy uses

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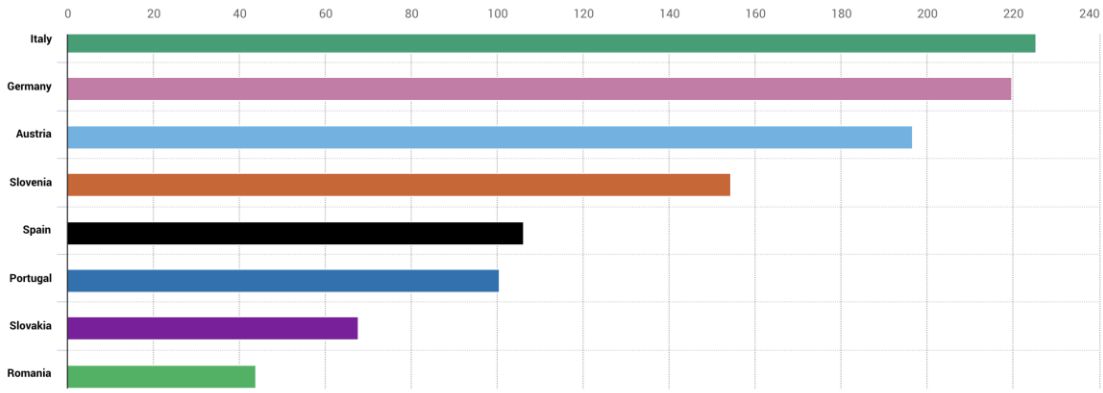

Figure 8 - Final energy consumption by agriculture/forestry per hectare of utilised agricultural area, in the 8 partners countries in 2022 [5].

4.1.4 Ammonia emissions from agriculture

The ammonia emissions from agriculture indicator is part of the EU Sustainable Development Goals (SDG) indicator set. It is used to monitor progress towards SDG on ending hunger and malnutrition, which is embedded in the European Commission's Priorities under the 'European Green Deal'.
SDG, among other things, seeks for sustainable and resilient food production systems. Ammonia (NH3) is a colourless, pungent-smelling and corrosive gas that is produced by the decay of organic vegetable matter and from the excrement of humans and animals. When released into the atmosphere, ammonia contributes to the level of air pollution. The agricultural sector continues to be the overwhelming contributor to ammonia emissions in the EU.
Measures implemented under the Common Agricultural Policy (CAP) and the Water Framework Directive has led to EU reductions of ammonia emissions from agriculture since the 1990s. Specifically, measures to change agricultural practices brought about reductions in the use of nitrogenous fertilisers and reduced livestock density per hectare across the EU. The National Emission Reduction Commitments Directive (NEC Directive) sets national emission reduction commitments for Member States and the EU for five important air pollutants, including ammonia.

The indicator measures the amount of ammonia (NH3) emissions as a result of the agricultural production. This comprises manure management, inorganic N-fertilizers and animal manure applied to soil as well as urine and dung deposited by grazing animals. Ammonia emissions per hectare are calculated using the total utilised agricultural area (UAA) of the relevant year as denominator.

In Figure 9 it can be observed 2001 data on Ammonia emissions from agriculture in the 8 partners countries, from Eurostat's 2021 annual crop statistics.

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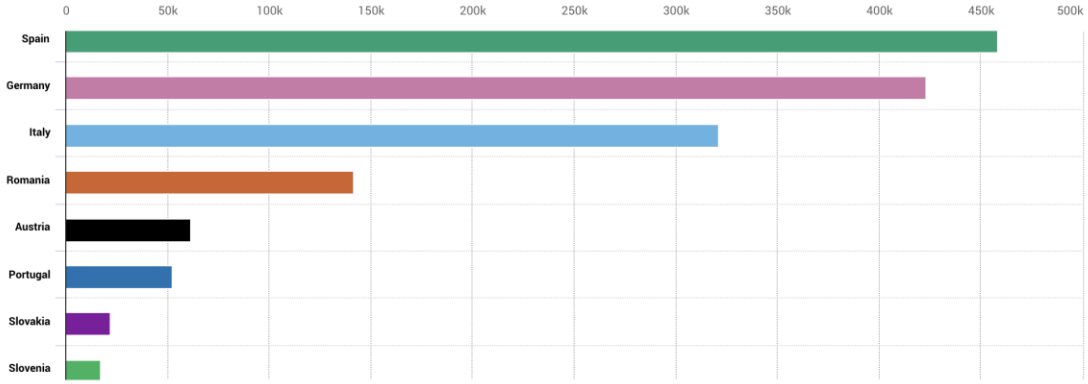

Figure 9. Ammonia emissions from agriculture in the 8 partners countries, in 2021 (Units: tonnes and kg per hectare utilized agricultural area (UAA) [5].

4.1.5 Greenhouse gas emissions from agriculture

This indicator tracks trends in greenhouse gas (GHG) emissions by agriculture, estimated and reported under the United Nations Framework Convention on Climate Change (UNFCCC), the Kyoto Protocol and the Decision 525/2013/EC.

The greenhouse gases presented in figure 10 are listed below:

- Greenhouse gases (CO2, N2O in CO2 equivalent, CH4 in CO2 equivalent, HFC in CO2 equivalent, PFC in CO2 equivalent, SF6 in CO2 equivalent, NF3 in CO2 equivalent)
- Carbon dioxide
- Methane
- Methane (CO2 equivalent)
- Nitrous oxide
- Nitrous oxide (CO2 equivalent)
- Hydrofluorocarbons (CO2 equivalent)
- Perfluorocarbons (CO2 equivalent)
- Hydrofluorocarbons and perfluorocarbons - not specified mix (CO2 equivalent)
- Sulphur hexafluoride (CO2 equivalent)
- Nitrogen trifluoride (CO2 equivalent)

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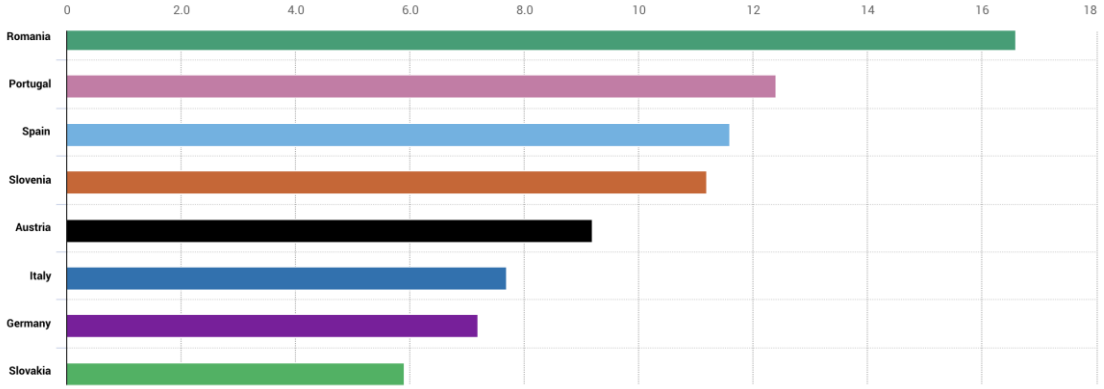

Figure 10. Total Greenhouse gas emissions from agriculture in the 8 partners countries, in 2021 [5].

Unlike emissions from burning fossil fuels, greenhouse gas emissions from agriculture are the product of intricate, difficult-to-measure natural processes.

- Enteric fermentation, the process by which livestock digest their dung, and livestock management practices are the main sources of methane. It is the main source of greenhouse gas emissions from agriculture.
- Nitrous oxide, which is typically produced by managing manure and applying agricultural fertiliser to soils, is the second biggest contributor.
- The increased breakdown of plant matter in soils and the conversion of land to agricultural use are the main sources of carbon dioxide emissions. The increasing amount of plant matter stored in farmland soils partially offsets those emissions.

Planting extra crops (sometimes referred to as cover crops) outside of the main growing season can help reduce carbon dioxide emissions. Reducing carbon dioxide emissions can also be achieved by using farming techniques that disturb the land less.
Limiting the amount of fertiliser applied and not applying when conditions are more conducive to nitrous oxide generation are two ways to reduce the formation of nitrous oxide.
By using manure management techniques that enable the absorption and use of methane emissions, such as an anaerobic digester, methane emissions from manure can be decreased [7].

4.1.6 Estimated soil loss by water erosion by land cover type.

Soil erosion by water is one of the most widespread forms of soil degradation in Europe. The indicator soil erosion estimates the soil loss by water erosion processes (rain splash, sheetwash and rills) for different land cover types which are based on the Corine Land

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Cover nomenclature: Agricultural areas and natural grassland, split in the latter two land cover types:

i) Agricultural areas excluding pastures.
ii) Pastures and natural grassland.

Figure 11 represents the affected area represented in tonnes per hectares per year. That affected area considers only potentially erosive-prone land, meaning that artificial, sandy, rocky and icy surfaces as well as wetlands and water bodies are not included in the area on which the indicator is based. Data on soil erosion are published under agreement with the data provider – Joint Research Centre of the European Commission (JRC – ISPRA). These data represent the soil erosion rates by water estimated on the basis of the Revised Universal Soil Loss Equation (RUSLE) empirical computer model.

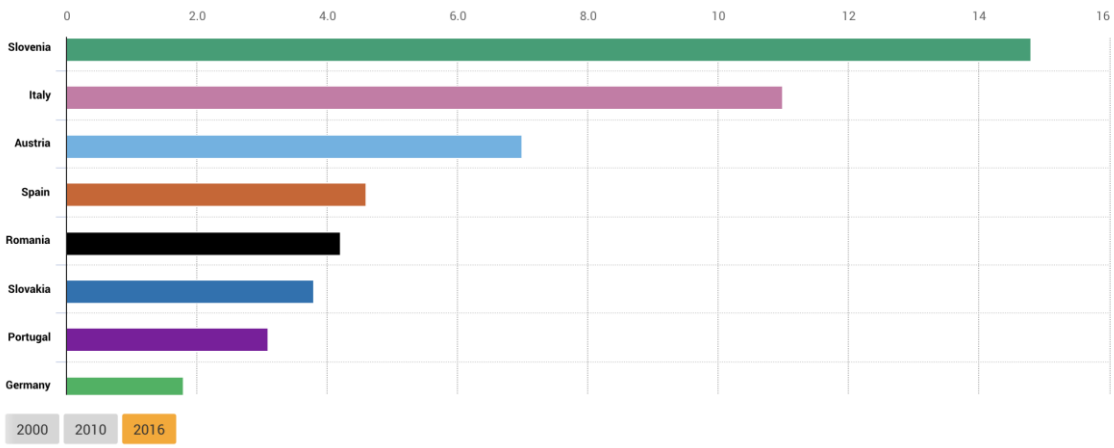

Figure 11. Estimated soil loss by water erosion by land cover type, Agricultural areas and natural grassland; Agricultural areas (excluding pastures) and Pastures and natural grassland in the 8 partners countries, in 2016 [5].

4.2 Waste

To promote and create a sustainable and circular economy by 2050, means reconsidering the entire life cycle of products as well as promoting sustainable consumption and the circular economy. This should lead to a reduction in the consumption of resources, less waste and fewer greenhouse gas emissions [8].

Higher Climate Benefits should be obtained through efficient waste Management policies, namely via an increase of recycling products and rates and CO2 emissions reduction for the waste streams. The EU's Circular Economy Action Plan includes measures about:

- Packaging and plastics
- Sustainable textiles

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- Electronics and ICT
- Construction and buildings
- Batteries and vehicles
- The food chain.
- Critical raw materials
- Repairing and reusing goods

4.2.1 Total waste

Although figure 12 represents the total waste, the results are also available in terms of Generation of waste by: waste category, hazardousness and a group that includes: Generation of waste, treatment of waste and waste treatment facilities (for obtain this information go to: [5]).
The data set on the generation of waste was broken down into waste categories and the source of waste generation; the data set on the treatment of waste was broken down into the type of treatment and waste category. The dataset on waste treatment facilities is broken down into the measurement variable (number of facilities, capacity).

Waste categories:
The data sets contain a breakdown into 51 waste categories according to the European Waste Classification for statistical purposes: EWC-Stat. It is a mainly substance-oriented classification, and it distinguishes hazardous and non-hazardous waste. The classification is linked to the administrative classification List of Wastes.

Source of waste generation:
The generation of waste is attributed to either production or consumption activities. The actor handing over the waste to the waste management system is regarded as the source. For production activities a further breakdown is supplied in 18 economic activities. Three of these activities are linked to the waste management and will contain secondary waste: Waste collection, treatment, and disposal activities; materials recovery, Remediation activities and other waste management services and Wholesale of waste and scrap. In addition to the waste generated by businesses waste is generated by households.

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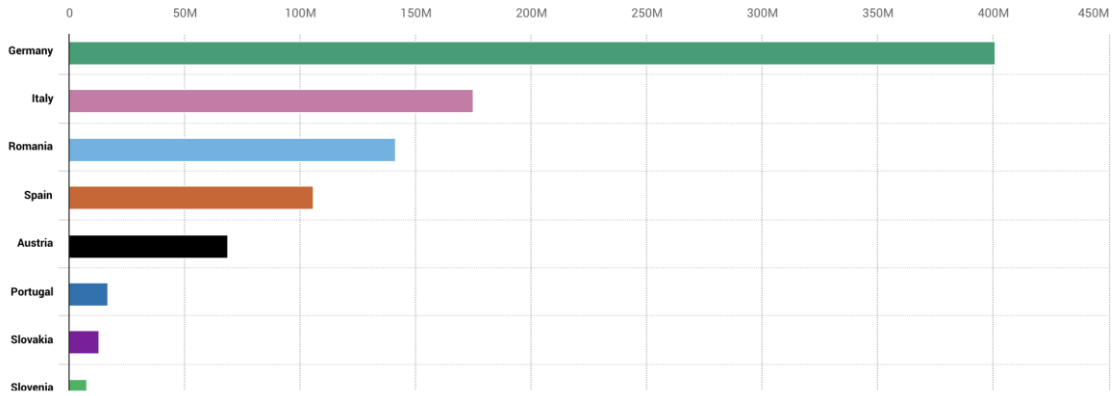

Figure 12. Total Generation of waste in the 8 partners countries, in 2020, (Unit of measure: Tonne) [5].

The share of different economic activities and of households in total waste generation in 2020 is presented in Figure 13. In the EU, construction contributed 37.5 % of the total in 2020 and was followed by mining and quarrying (23.4 %), waste and water services (10.8 %), manufacturing (10.6 %) and households (9.4 %); the remaining 8.2 % was waste generated from other economic activities, mainly services (4.4 %) and energy (2.3 %).

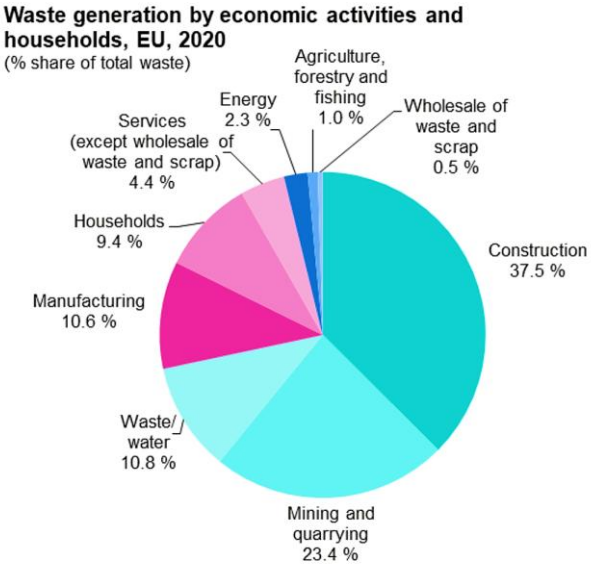

Figure 13. Waste generation by economic activities and households, EU, 2020 (% share of total waste) [5].

4.2.2 Plastic waste

In Figure 14 it can be seen the amount of plastic packaging waste generation per capita in the 8 partners countries, in 2020. This indicator includes plastic packaging waste. 'Packaging' in this context means all products made of any materials of any nature to be used for the containment, protection, handling, delivery and presentation of goods, from raw materials to processed goods, from the producer to the user or the consumer. 'Non-returnable' items used for the same purposes shall also be considered to constitute packaging. 'Packaging waste' means any packaging or packaging material covered by the

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definition of waste in the Waste Framework Directive 2008/98/EC, excluding production residues (Art.3(1): 'waste' means any substance or object which the holder discards or intends or is required to discard).

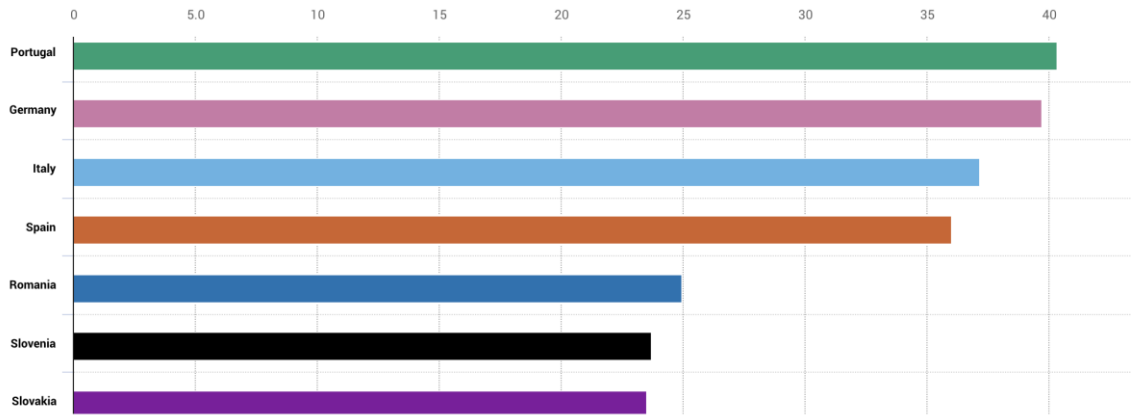

Figure 14 Generation of plastic packaging waste per capita in the 8 partners countries, in 2020 [5].

Specific institutions and localities around the world are working to reduce plastic waste. For instance, the sale of plastic water bottles is prohibited at the San Francisco International Airport. Passengers can only buy refillable glass or metal bottles at the airport beginning in August 2019. In 2018, the European Commission released the Single-Use Plastics (SUP) Directive, which outlines a plan for plastics in a circular economy [9].

4.2.3 Waste treatment

In 2020, some 1 971 million tonnes of waste were treated in the EU. This does not include exported waste but includes the treatment of waste imported into the EU. The reported amounts are therefore not directly comparable with those on waste generation [5].

Figure 15 shows the total waste treatment in the EU, in the 8, partners countries, in 2020.

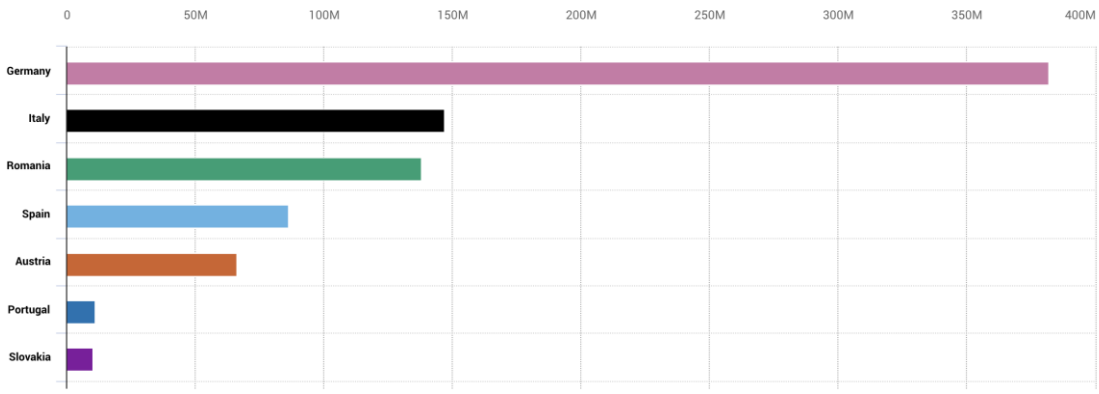

Figure 15 Treatment of waste, hazardous and non-hazardous – Total, in the 8, partners countries, in 2020 [5].

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4.2.3.1 Waste management operations

In figures 16 to 20 it can be observed the amounts of treatment of waste, by waste management operations, in the 8, partners countries, in 2020.

Based on the treatment operations defined in Directive 2008/98/EC a distinction is made in treatment types:

- Recovery -Recycling and backfilling (excluding energy recovery)
- Energy recovery
- Recovery - backfilling
- Recovery - recycling
- Incineration (without energy recovery)
- Disposal - landfill
- Disposal - other
- Landfill / disposal

The quantity of waste recovered, in other words recycled, used for backfilling (the use of waste in excavated areas for the purpose of slope reclamation or safety or for engineering purposes in landscaping) or incinerated with energy recovery increased by 33.9 % from 870 million tonnes in 2004 to 1 165 million tonnes in 2020; as a result, the share of such recovery in total waste treatment rose from 45.9 % in 2004 to 59.1 % in 2020. The quantity of waste subject to disposal decreased from 1 027 million tonnes in 2004 to 806 million tonnes in 2020, which was a decrease of 21.5 %. The share of disposal in total waste treatment decreased from 54.1 % in 2004 to 40.9 % in 2020.
As stated above, in the EU in 2020, more than a half (59.1 %) of the waste was treated in recovery operations: recycling (39.9 % of the total treated waste), backfilling (12.7 %) or energy recovery (6.5 %). The remaining 40.9 % was either landfilled (32.2 %), incinerated without energy recovery (0.5 %) or disposed of otherwise (8.2 %). Significant differences could be observed among the EU Member States regarding the use they made of these various treatment methods. For instance, some Member States had very high recycling rates (For instance Italy, Belgium, Slovakia and Latvia), in others landfill is the prevailing treatment category (Romania, Bulgaria, Finland, Sweden and Greece).

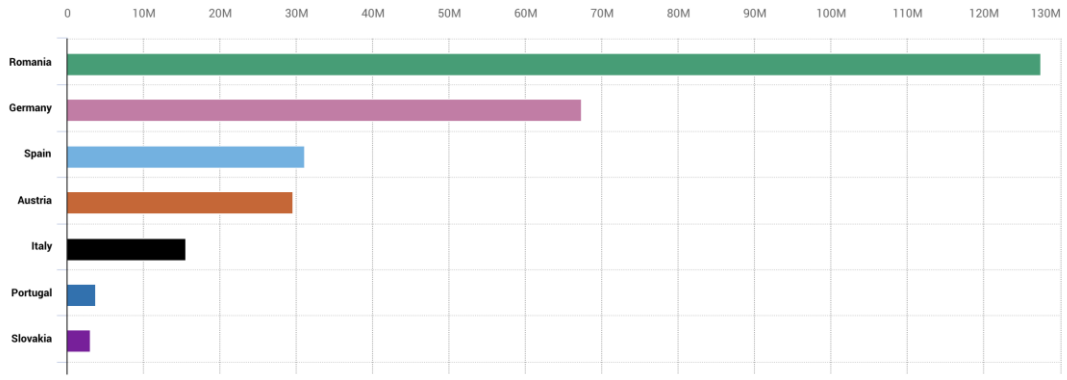

Figure 16 – Total Amount of waste that is Disposal-landfilled in the 8 partners countries in 2020 [5].

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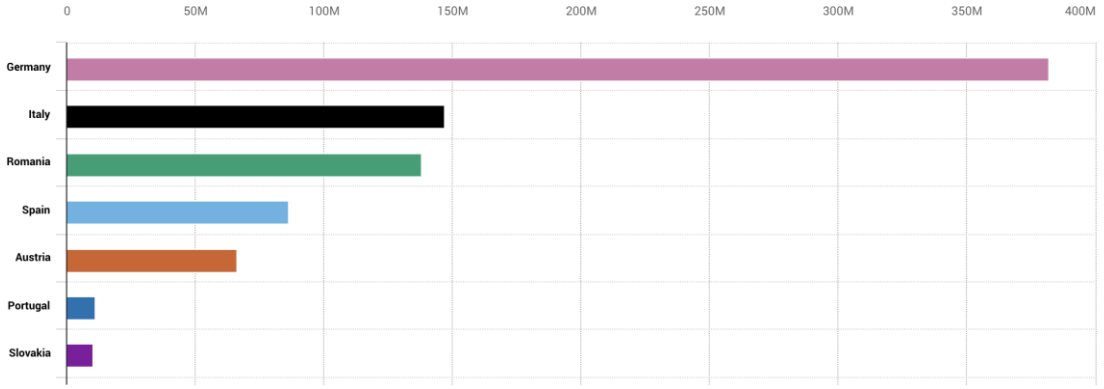

Figure 17- Amount of waste that is Disposal-incinerated in the 8 partners countries in 2020 [5].

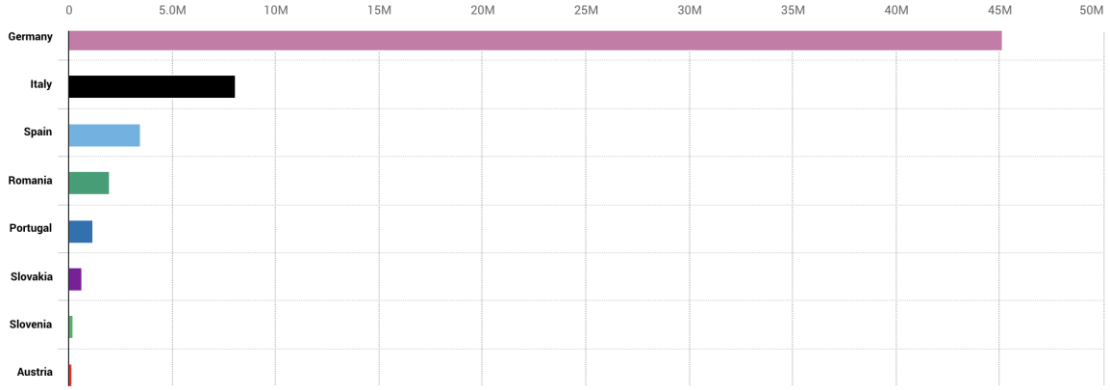

Figure 18- Amount of waste that is recovered-energy recovered in the 8 partners countries in 2020 [5].

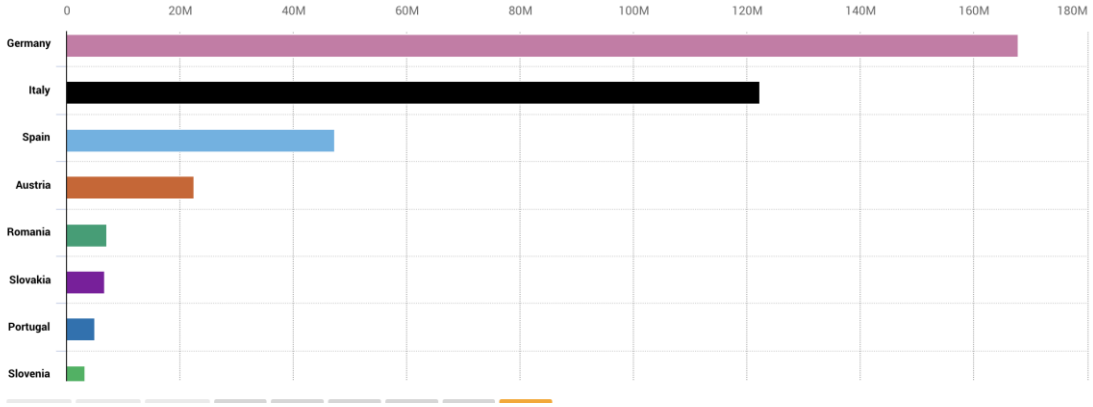

Figure 19- Amount of waste that is recovery-recycled in the 8 partners countries in 2020 [5].

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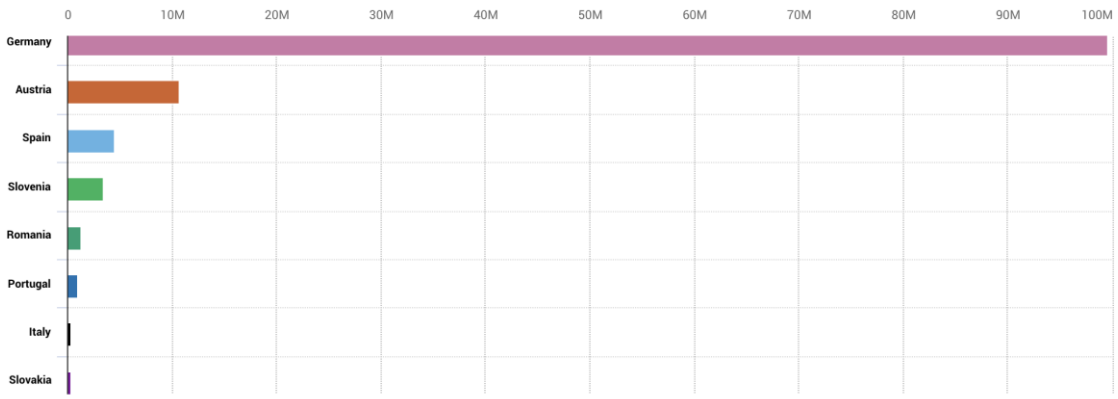

Figure 20 - Amount of waste that is recovery- backfilling in the 8 partners countries in 2020 [5].

4.3 Transports

Transport represents almost a quarter of Europe's greenhouse gas emissions and is the main cause of air pollution in cities.

4.3.1 Final energy consumption in transport by type of fuel

Final energy consumption in transport covers the energy consumption of the following transport modes: road transport (cars, buses, trucks, etc), rail transport (trains, metro, trams, etc.), domestic aviation, domestic navigation and pipeline transport. Quantities delivered to international aviation and international marine bunkers are excluded from the final energy consumption. Energy used at airports, train stations, ports and bus stations is excluded from transport and included in the service sector. Off road use in agriculture and forestry is excluded from transport and included in the final energy consumption of agriculture and forestry. Fuel used in ships for fishing is excluded from transport and included in in the final energy consumption of the fishing sector.

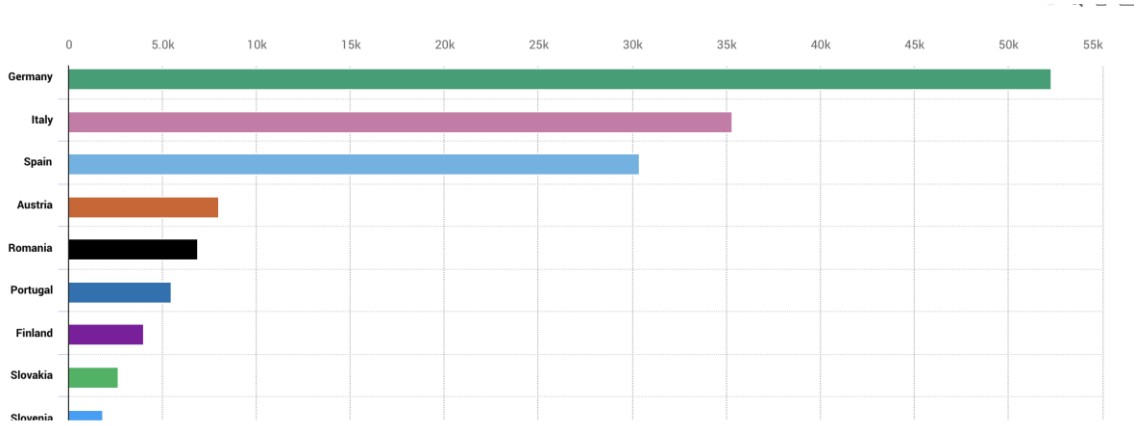

Figure 21 – Final energy consumption in transporting- all fuels- in the 8 partners countries in 2021. Unit of measure: Thousand tonnes of oil equivalent [5].

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From those totals the consumption of electricity in transportation is represented in figure 22

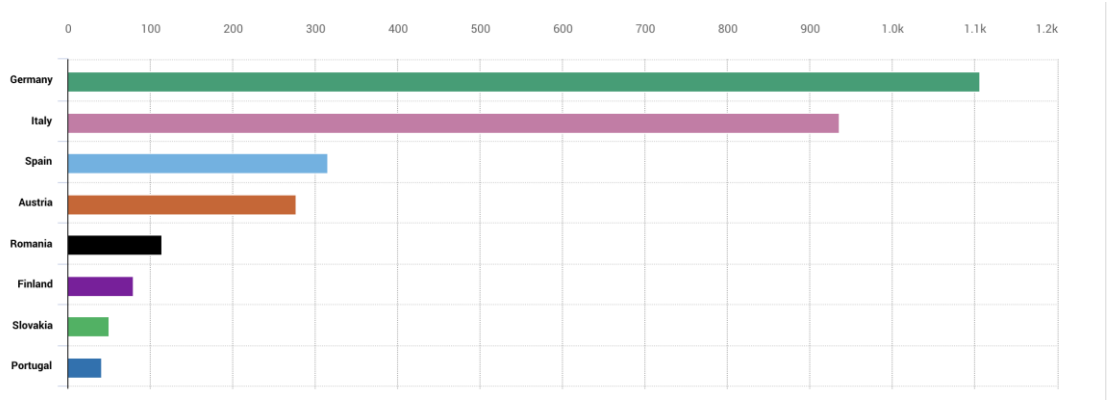

Figure 22 – Final Electrical energy consumption in transporting in the 8 partners countries in 2021; Unit of measure: Thousand tonnes of oil equivalent [5].

4.3.2 Policies and measures by transport sector

4.3.2.1 Road transport

Europe's greenhouse gas emissions and is the main cause of air pollution in cities. In 2020, road transport contributed to 24% of the EU's total emissions of carbon dioxide (CO2), the main greenhouse gas. To reduce Reducing CO_2 emissions from vehicles the following meaures should be taken [10]:

- Light-duty vehicles: passenger cars and vans, produce around 15% of the EU's CO2 emissions. Regulation (EU) 2019/631 sets binding CO2 emission targets for the new car and van fleets, which were updated in April 2023 to bring them in line with the EU's 2030 and 2050 climate objectives.

- CO_2 labelling of cars: To help drivers choose new cars with low fuel consumption, EU legislation requires Member States to ensure that relevant information is provided to consumers, including a label showing a car's fuel efficiency and CO2 emissions.

- Heavy-duty vehicles: trucks and buses are responsible for more than a quarter of CO2 emissions from road transport in the EU, and for 6% of total EU greenhouse gas emissions. On 20 June 2019, the European Parliament and the Council adopted a Regulation setting CO2 emission standards for heavy-duty vehicles, with targets for reducing the average emissions from new lorries for 2025 and 2030.
- Fuel quality: Fuel quality is an important element in reducing greenhouse gas emissions from transport. EU legislation requires the greenhouse gas intensity of vehicle fuels to be cut by up to 6% by 2020.

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4.3.2.2 Maritime transports

Reducing emissions from the shipping sector: International shipping is a large and growing source of greenhouse gas emissions. We support global action to tackle these emissions.

While maritime transport plays an essential role in the EU economy and is one of the most energy-efficient modes of transport, it is also a large and growing source of greenhouse gas emissions. In 2018, global shipping emissions represented 1 076 million tonnes of CO2, and were responsible for around 2.9% of global emissions caused by human activities.

As part of the European Commission's legislative proposals to deliver the European Green Deal - the 'Fit for 55' package - published on 14 July 2021, several proposals addressed maritime transport's climate impact, in addition to the extension of the EU ETS.
This includes:

- A new FuelEU maritime Regulation to boost the demand for marine renewable and low-carbon fuels, by setting a maximum limit on the greenhouse gas content of energy used by ships calling at European ports and by encouraging zero-emission technology at berth (where ships stay in ports), with a technology-neutral approach;
- Revising the Directive on Deployment of Alternative Fuels Infrastructure, which would set, among other benefits, mandatory targets for shore-side electricity supply at maritime and inland waterway ports;
- Accelerating the supply of renewables in the EU, through a revision of the Renewable Energy Directive (RED), which increases the current EU target of at least 32% of renewable energy sources in the overall energy mix to at least 40% by 2030, with a focus on sectors where progress has been slower to date – including transport;
- Revising the existing Energy Taxation Directive (ETD), which aims to align the taxation of energy products with the EU's climate objectives and remove outdated exemptions, such as those for the intra-EU maritime transport sector.

This basket of measures reflects the EU's goal to cut greenhouse gas emissions by addressing the various barriers to the decarbonisation of the shipping sector (technological barriers, economic barriers, etc.). The Commission aims to do that through two complementary angles: first, the improvement of energy efficiency (i.e. using less fuel) and, second, the greater use of renewable and low-carbon fuels (i.e. using cleaner fuels). These measures will allow the creation of a virtuous ecosystem for such cleaner fuels, as they will boost at the same time fuel demand, distribution, and supply [11].

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4.3.2.3 Aviation

Aviation is one of the fastest-growing sources of greenhouse gas emissions. The EU is taking action to reduce aviation emissions in Europe and working with the international community to develop measures with global reach.
Reducing emissions from aviation: Aviation is one of the fastest-growing sources of greenhouse gas emissions. We want to reduce aviation emissions in Europe and beyond.

On 14 July 2021, the European Commission adopted a series of legislative proposals setting out how it intends to achieve climate neutrality in the EU by 2050, including the intermediate target of an at least 55% net reduction in greenhouse gas emissions by 2030. The package proposes to revise several pieces of EU climate legislation, including the EU ETS, Effort Sharing Regulation, transport and land use legislation, setting out in real terms the ways in which the Commission intends to reach EU climate targets under the European Green Deal.

Airlines will be required to

- monitor emissions on all international routes;
- offset emissions from routes included in the scheme by purchasing eligible emission units generated by projects that reduce emissions in other sectors (e.g. renewable energy).

During the period 2021-2035, and based on expected participation, the scheme is estimated to offset around 80% of the emissions above 2020 levels. This is because participation in the first phases is voluntary for states, and there are exemptions for those with low aviation activity. All EU countries will join the scheme from the start [12].

4.4 Energy production

4.4.1 Energy efficiency

This Parameter covers indicators for monitoring progress towards energy efficiency targets implemented by Directive 2012/27/EU on energy efficiency and Directive (EU) 2023/1791 on energy efficiency (recast). The indicator measures the level of energy consumption and distance to 2020/2030 targets. The energy saving could improve the security of the energy supply by reducing the dependence on fuel imports. The improvement in energy efficiency also improves the competitiveness of European industry and services; for households it reduces the energy bill. Energy saving could also contribute to reducing the GHG emissions from fuel combustion.
Energy consumptions are measured in Mtoe.

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The distance to target in absolute terms (Mtoe) is calculated as a difference between the observed energy consumption in a given year and the absolute primary and final energy consumption targets in 2020 according to Directive 2012/27/EU and in 2030 according to Directive (EU) 2023/1791.

The distance to target in relative terms (as percentage) is calculated as a ratio of the distance to target in absolute terms in a given year to primary and final energy consumption targets in 2020 according to Directive 2012/27/EU and in 2030 according to Directive (EU) 2023/1791.

Basic data on energy quantities are given in fuel specific units e.g. solid and liquid fuels in thousand tonnes, electricity in gigawatt-hours, heat and gases in terajoules. The basic energy quantities data are converted to energy units, i.e. to terajoules and tonnes of oil equivalent to allow comparison of different fuel types.
Energy efficiency indicators included in this dataset are using these units:

- 1 000 000 tonnes of oil equivalent (Mtoe)
- Index with base year 2005 = 100
- Percentage (%)

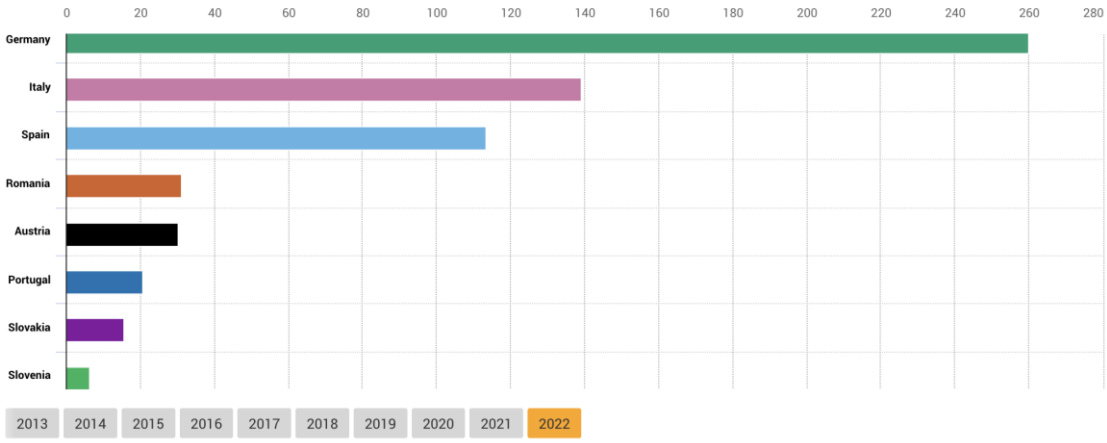

Figure 23 – Total energy efficiency- Primary energy consumption in the 8 partners countries in 2022. Unit of measure: Million tonnes of oil equivalent [5].

4.4.2 Energy production

Measured by the Quantities for crude oil, petroleum products, natural gas and manufactures gases, electricity and derived heat, solid fossil fuels, renewables and wastes covering the full spectrum of the energy sector from supply through transformation to final consumption by sector and fuel type. Also, annual imports and exports data of various energy carriers by country of origin and destination, as well as infrastructure information.

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In Figure 24 it can be seen the gross electricity production with combined heat and power units and with all Generation technologies in the 8 partners countries in 2021

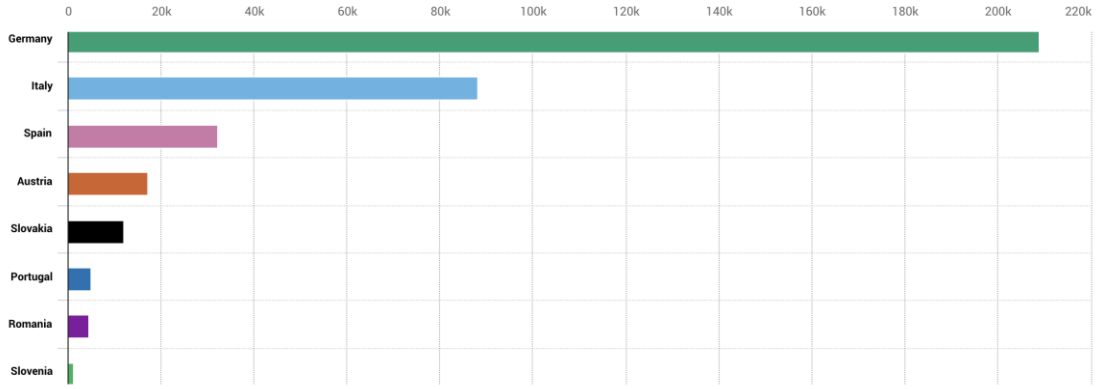

Figure 24 – Gross electricity production with combined heat and power units and with all Generation technologies in the 8 partners countries in 2021. unit of measure: Gigawatt-hour [5].

4.4.3 Renewable energy

The Renewable Energy Directive establishes common rules and targets for the development of renewable energy across all sectors of the economy.
The revised Renewable Energy Directive, adopted in 2023, raises the EU's binding renewable energy target for 2030 to a minimum of 42.5%. It means almost doubling the existing share of renewable energy in the EU [13].

In figure 25 it can be observed the percentage of renewable energy in gross final energy consumption, according to the Renewable Energy Directive. In figures 26 to 28, similar results are obtained, nut shared by sector. The gross final energy consumption is the energy used by end-consumers (final energy consumption) plus grid losses and self-consumption of power plants.

This indicator is part of the EU Sustainable Development Goals (SDG) indicator set. It is used to monitor progress on affordable and clean energy and on climate action, which are embedded in the European commission's Priorities under the 'European Green Deal'. SDG calls for ensuring universal access to modern energy services, improving energy efficiency and increasing the share of renewable energy. To accelerate the transition to an affordable, reliable, and sustainable energy system, countries need to facilitate access to clean energy research, promote investment in energy infrastructure and clean energy technology. SDG also aims to implement the commitment to the United Nations Framework Convention on Climate Change and operationalise the Green Climate Fund.

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This indicator can be considered as identical to global SDG indicator "Renewable energy share in the total final energy consumption".

The EU is implementing an interconnected set of measures "Fit for 55" to reach the new climate targets for 2030 and has followed a new strategic plan REPowerEU to reduce the EU dependency on energy imports. The European Green Deal envisages the decarbonisation of Europe's energy systems in order to reach climate neutrality by 2050. For this purpose, the EU will turn towards a fully integrated, interconnected and digitalised renewable energy system, while at the same time ensuring security of supply at affordable prices. Moreover, EU cohesion policy funds (2021–2027) including the 'Next Generation EU' recovery plan provide funding for investments in renewable energies. The revised Renewable energy Directives set an ambitious EU target to be reached by 2030.

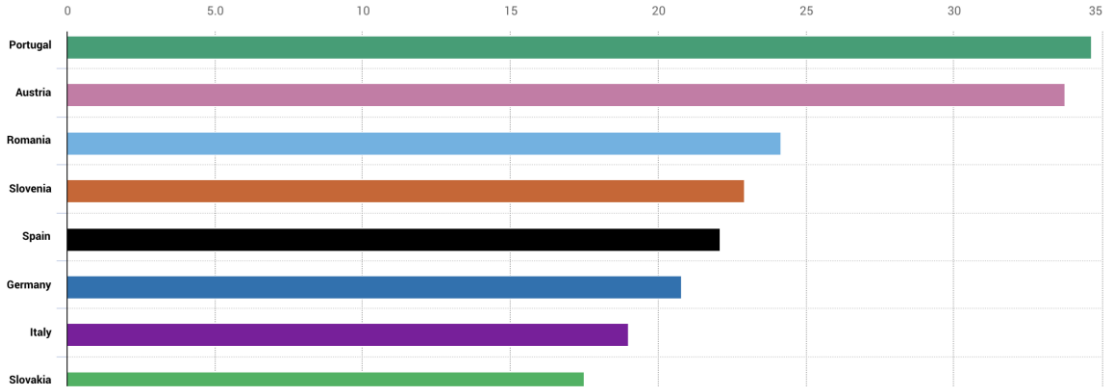

Figure 25 – percentage of renewable energy in gross final energy consumption in the 8 partners countries in 2022 [5].

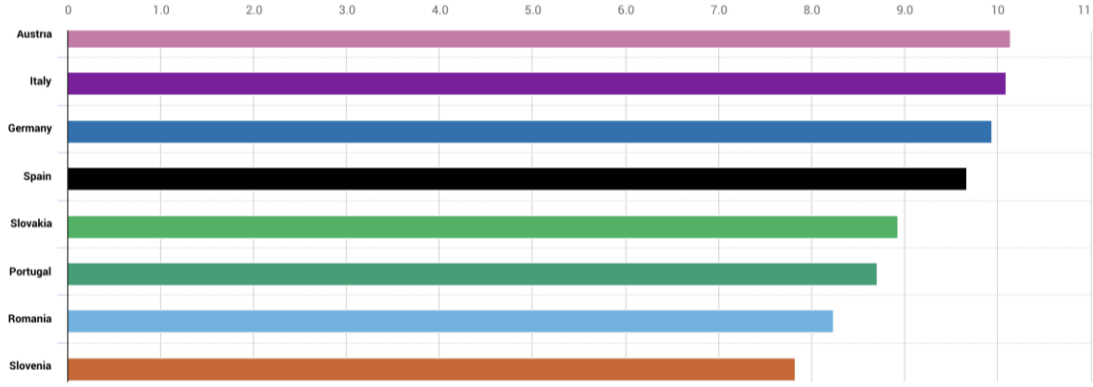

Figure 26 – Share of renewable energy in gross final energy consumption in transport in the 8 partners countries in 2022 [5].

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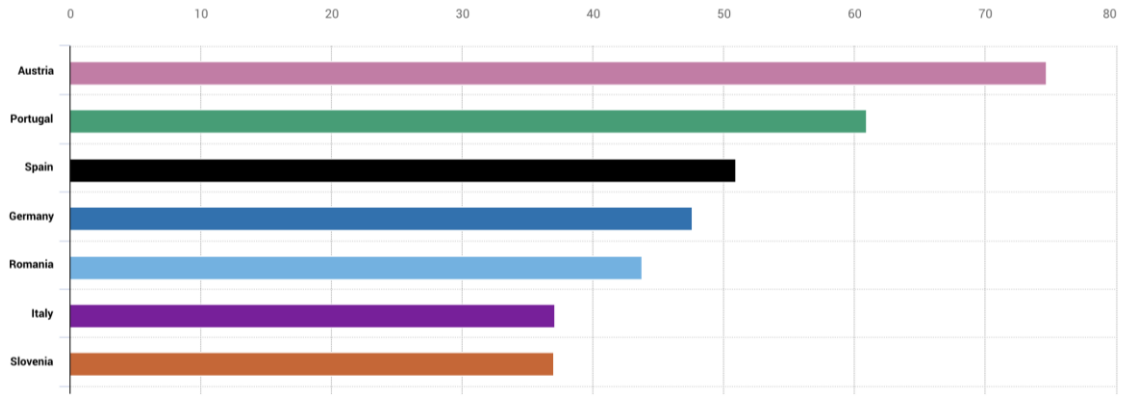

Figure 27 – Share of renewable energy in gross final energy consumption in electricity in the 8 partners countries in 2022 [5].

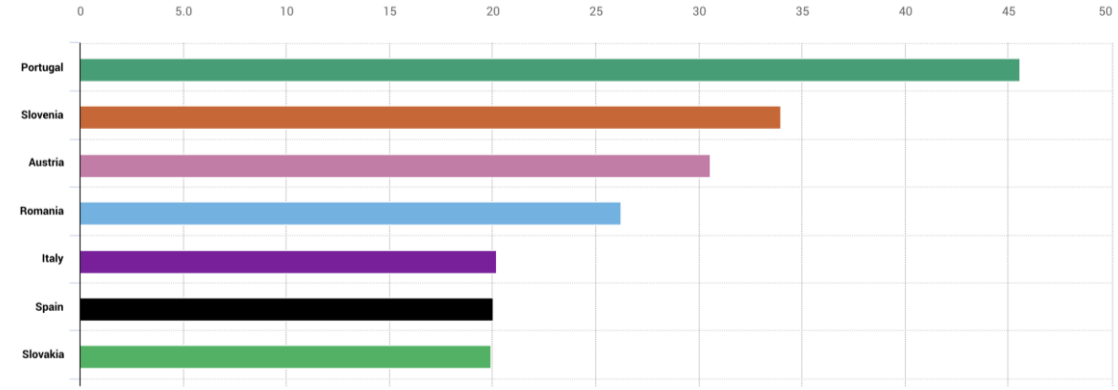

Figure 28 - Share of renewable energy in gross final energy consumption in heating and cooling in the 8 partners countries in 2022 [5].

4.5 Residential activities

The buildings sector is a key contributor to greenhouse gas (GHG) emissions, representing 35% of energy-related EU emissions in 2021. These emissions result partly from the direct use of fossil fuels in buildings (e.g., oil and gas used in boilers for heating) and partly from the production of electricity and heat for use in buildings (e.g., electricity consumed by water heaters, lighting, electrical devices, cooling systems, etc.).

The European Green Deal, the Renovation Wave Initiative, and the EU's recovery plan place a strong emphasis on reductions in GHG emissions and energy use from buildings. The REPowerEU plan published by the European Commission in May 2022, which aims to end the EU's dependence on Russian fossil fuels while tackling the climate crisis, calls for additional savings and energy efficiency gains in buildings, particularly through the Energy Performance of Buildings Directive (currently under negotiation). It also introduces the European Solar Rooftop Initiative with a proposal for a legally binding EU solar rooftop obligation for certain categories of buildings.

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Improvements to buildings, such as better insulation and decarbonised heating and cooling systems, can help reduce emissions from fossil fuel use. While often providing a lower-emitting alternative, heating systems such as heat pumps increase a building's use of electricity. Unless this electricity demand is met by the production of renewable or decarbonised energy, this can lead to a sectoral shift in emissions from the buildings to the electricity sector.

In figure 29 it can be observed the percentage exchange from 2005-2021 of greenhouse gas emissions from energy use in buildings by the 8 partners countries. Historical and projected emissions from the use of fossil fuels in buildings differ significantly from one Member State to another. Lithuania, Luxembourg, Malta and Romania are the only Member States where emissions increased over the period 2005-2021. In all other Member States, emissions decreased, with Denmark, Greece, Slovenia and Sweden reporting the largest reductions (from 54% to 64%) and Luxembourg, Romania and Lithuania reporting the lowest decrease (between 1% and 10%).

Compared to 2005, national projections of emissions from buildings foresee a further decrease in emissions by 2030 in all but two Member States: Malta, for which the increase correlates to increased electricity use for air conditioning; and Romania because of projected higher income levels. At the same time, additional measures in eight countries (Denmark, Finland, Germany, Greece, Ireland, Portugal, Slovenia and Sweden) are expected to result in reducing emissions from buildings by 55% or more between 2005 and 2030.

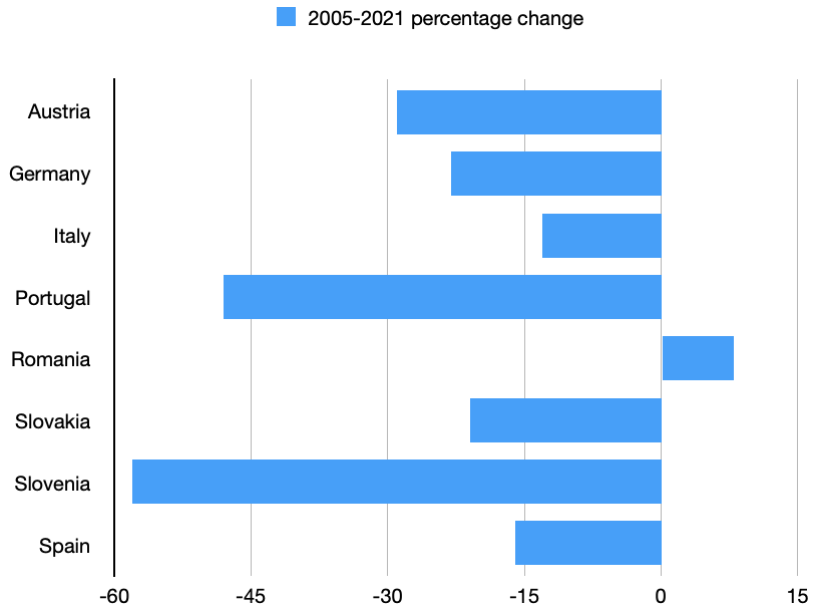

Figure 29. Percentage exchange from 2005-2021 of greenhouse gas emissions from energy use in buildings by the 8 partners countries [5].

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SDG aims to renew and plan cities and other human settlements in a way that offers opportunities for all, with access to basic services, energy, housing, transportation and green public spaces, while reducing resource use and environmental impact. Monitoring SDG in an EU context means looking at developments in the quality of life in cities and communities, sustainable mobility and adverse environmental impacts [14].

4.5.1 Quality of life in cities and communities Severe housing deprivation rate

This indicator is defined as the percentage of the population living in a dwelling which is considered as overcrowded, while also exhibiting at least one of the following housing deprivation measures: (i) a leaking roof, (ii) no bath/shower and no indoor toilet, and (iii) considered too dark.
Population living in households suffering from noise This indicator measures the share of the population who declare they are affected by noise either from neighbours or from the street.

4.5.2 Settlement area per capita

This indicator captures the amount of settlement area such as for buildings, industrial and commercial areas, infrastructure and sports grounds, and includes both sealed and non-sealed surface

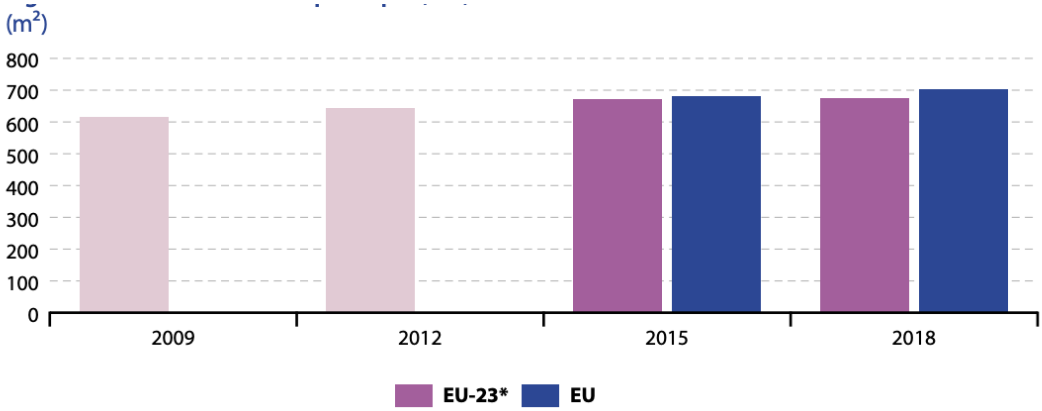

Figure 30. settlement area, m^2, per capita, EU, between 2009-2018 [1].

4.5.3 Recycling rate of municipal waste

This indicator measures the percentage of recycled municipal waste. Recycling includes material recycling, preparing for re-use, composting and anaerobic digestion. Figure 11.6: Recycling rate of municipal waste, EU, 2010–2021 (% of total municipal waste generate

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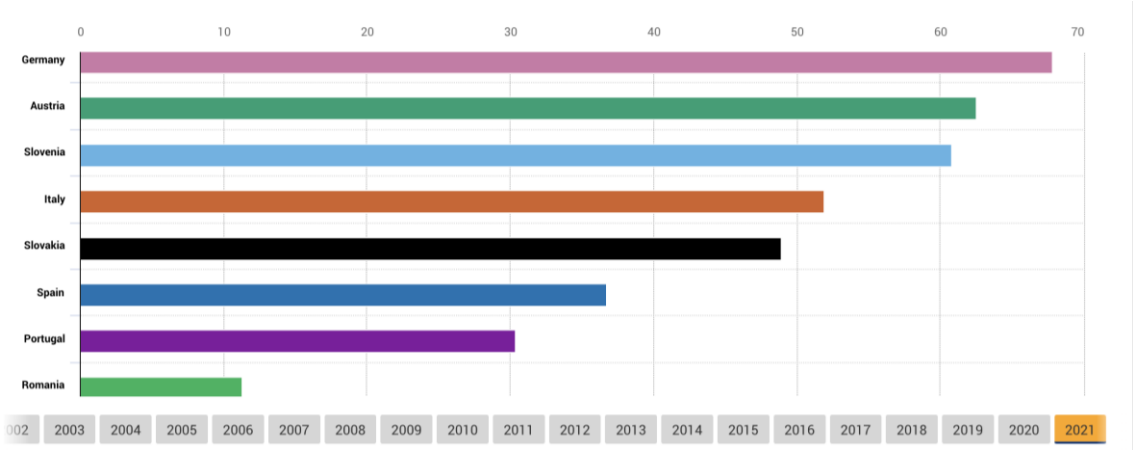

Figure 31. Recycling rate of municipal waste, in %, 2021, by the 8 partners countries [5].

4.5.4 Responsible consumption and production

SDG 12 calls for a comprehensive set of actions from businesses, policymakers and consumers to adapt to sustainable practices. It envisions sustainable production and consumption based on advanced technological capacity, resource efficiency and reduced global waste. Monitoring SDG 12 in an EU context focuses on developments in the areas of decoupling environmental pressures from economic growth, the green economy, and waste generation and management. 12.1 – Decoupling environmental pressures from economic growth Material footprint. The material footprint of the EU, also referred to as raw material consumption (RMC), quantifies the global extraction of materials needed to satisfy consumption of goods and services within the EU. For comparability reasons, raw material equivalents are estimated for imports and exports. Figure 12.1: Raw material consumption, EU, 2010–2020 (billion tonnes)

4.5.5 Raw material consumption (RMC)

The material footprint, also referred to as raw material consumption (RMC), represents the global demand for the extraction of materials (minerals, metal ore, biomass, fossil energy materials) induced by consumption of goods and services within a geographical reference area. Data for material footprints stem from material flow accounts, which model the flows of natural resources from the environment into the economy. They include domestic extraction of materials measured in tonnes of gross material (for example, gross ore or gross harvest) as well as imports and exports measured by estimates of the raw material equivalents of the products traded (domestic and abroad extraction required to produce the traded products). RMC thus shows the amount of extraction needed to produce the goods demanded by final users in the geographical reference area, irrespective of where in the world the material extraction took place.

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In Figure 32 it can be seen the raw material consumption in 2022, by the 8 ANGIE partners countries.

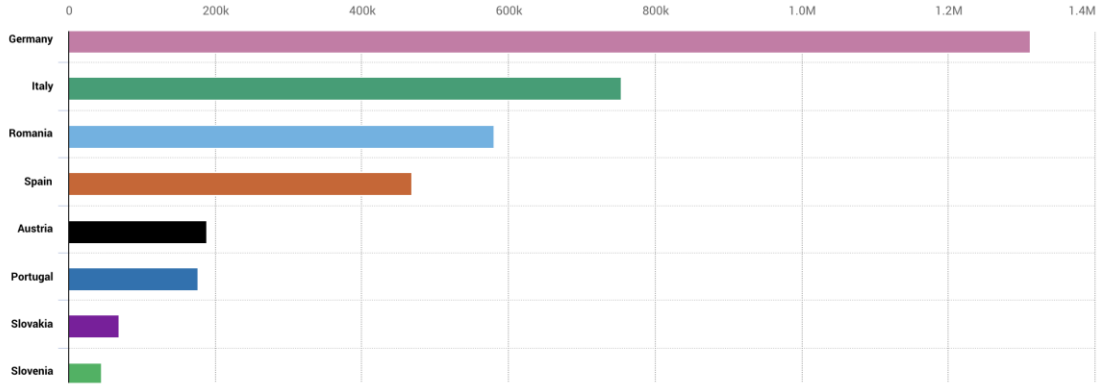

Figure. 32 Raw material consumption 2022, by the 8 partners countries, unit measure: thousand tonnes [5].

4.6 Industrial activities

The European Green Deal will have to foster major shifts in the European industrial structure, including transitions from fossil fuels to renewable energy and from combustion engine cars to electric cars. Shifting economies from brown to green would be a major, historic socio-economic transformation. Industry accounts for 25 percent of gross domestic product (GDP a monetary measure of the value of all final goods and services produced in a period.)
The industrial sector consists of three main categories of activity:

- Mining and quarrying
- Manufacturing, and
- Electricity, gas and water supply

Industrial activities have a significant contribution to the GDP of the European Union (EU), currently accounting for 24.3%3. The industrial share of GDP for individual Member States.

Numerous green industrial policy initiatives, implemented at the regional, national, and EU levels, are what define Europe. These projects may even be at odds with one another and are typically not coordinated. This is a serious problem since widely disparate green industrial policies across EU member states could distort fair competition and split the EU single market. Innovative cleantech businesses in Europe are unable to grow to the same extent as their US and Chinese competitors due to the fragmented EU single market for green technologies. Thus, creating a strong regulatory framework and enforcing competition laws are essential to guaranteeing access to a truly single, competitive EU market with uniform environmental standards. Put otherwise, the EU single market is a valuable resource for the development of an EU green industrial policy. Furthermore, in

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order to internalise externalities and take advantage of synergies and economies of scale, European nations and businesses would substantially profit from coordinated, cooperative efforts in several green technologies.
Such coordinated actions should learn from past failures, such as Germany's unsuccessful attempts at developing a solar panel industry despite generous solar energy subsidies.

It is often said that the European Green Deal must turn decarbonisation into an opportunity to revitalise the European economy, and thus to ensure long-term economic growth and jobs. That is, while heading towards climate neutrality by 2050, the European economy has to remain highly competitive at global level, in the context of increasing competition from other major economies.
New industrial policy' (NIP), stress that industrial policy should have aims beyond short-term competitiveness and economic growth. It should have a broader multi-dimensional objective, which can be captured in the notion of long-term social welfare. While the goal of climate policy is decarbonisation and the goal of (new) industrial policy is social welfare, green industrial policy must reconcile the goals of decarbonising the economy (like climate policy) and social welfare (like industrial policy) (Figure 33). Green industrial policy can then be defined as an industrial policy in which climate change mitigation becomes a binding constraint in achieving the social welfare goal. This combination of objectives immediately identifies the challenge of green industrial policy, which is to meet both goals simultaneously [15].

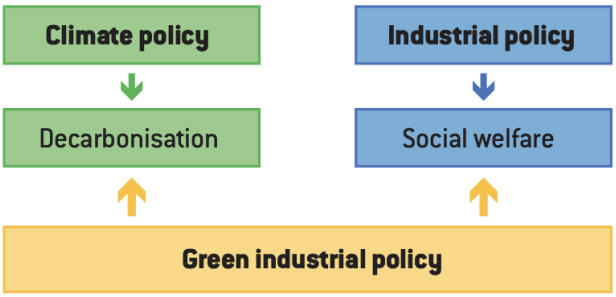

Figure. 33. Schematic representation of New Industrial policies: climate policy is decarbonisation and the goal of (new) industrial policy is social welfare, green industrial policy must reconcile the goals of decarbonising the economy (like climate policy) and social welfare (like industrial policy) [15].

4.6.1 Air emissions intensity of industry

This indicator measures the emissions intensity of particulate matter (PM10 and PM2.5) from the manufacturing sector. Air emissions are defined as flows of gaseous and particulate materials emitted into the atmosphere. Fine and coarse particulates (PM10) are less than 10 micrometres in diameter and can be carried deep into the lungs, where they can cause inflammation and exacerbate the condition of people suffering from heart

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and lung diseases. Fine particulates (PM2.5) are less than 2.5 micrometres in diameter and are therefore a subset of the PM10 particles. Their negative health impacts are more serious than PM10 because they can be drawn further into the lungs and may be more toxic. Emission intensity is calculated by dividing the sector's PM emissions by its gross value added (GVA), which is defined as output (at basic prices) minus intermediate consumption (at purchaser prices). In figures 34 and 35 it can be observed the Air emission intensity from industry in 2021, for the ANGIE partners countries.

Figure 34 - Air emission intensity from industry in 2021, for particle lower than 2.5µm, in the 8 partners countries; Grams per euro [5].

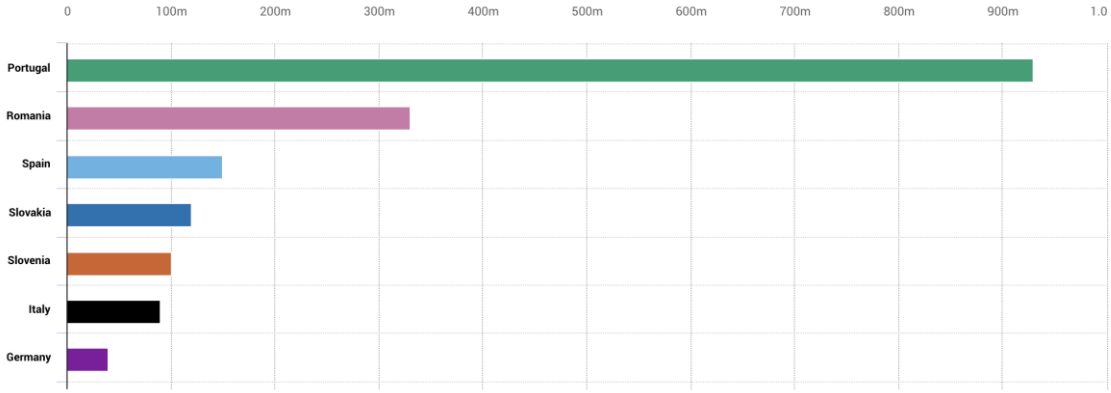

Figure 35 - Air emission intensity from industry, in the 8 partners countries in 2021, for particle lower than 10 µm; Grams per euro [5].

4.6.2 Gross value added of the environmental goods and services sector.

Environmental economy activities are expected to become more and more important at the global level. A recent report estimates that the global market volume for environmental technology and resource efficiency activities will increase by 7.3% per year until 2030. The increasing opportunities for the environmental economy, particularly for economic sectors that contribute to achieving net-zero emissions, are also highlighted in the International Energy Agency reports 'World energy outlook 2022' and 'Energy technology perspectives 2023' [16].

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The environmental goods and services sector (EGSS) is defined as that part of a country's economy that is engaged in producing goods and services that are used in environmental protection and resource management activities either domestically or abroad. Gross value added in EGSS represents the contribution of the environmental goods and services sector to GDP and is defined as the difference between the value of the sector's output and intermediate consumption. In figure 36 it can be seen the Gross value added of the environmental goods and services sector by the 8 partners countries, in 2020.

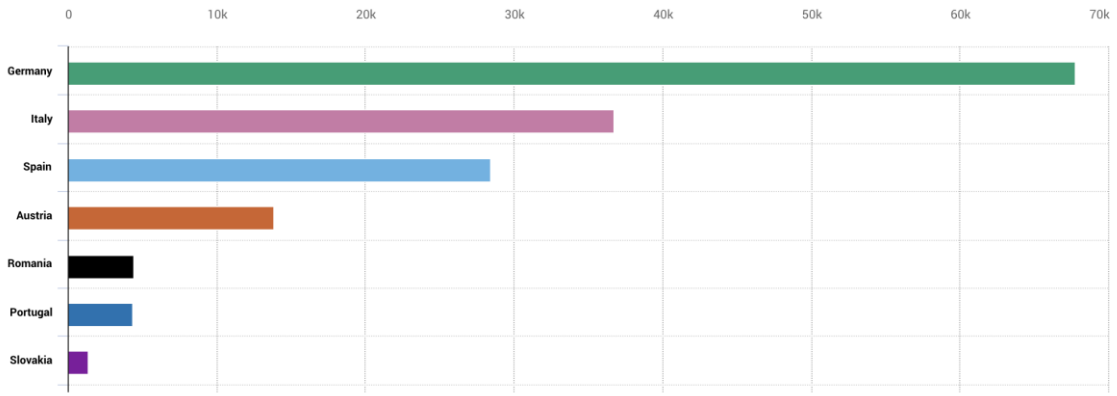

Figure 36 - Gross value added of the environmental goods and services sector by the 8 partners countries, in 2020, measure unit Million euro [5].

Figure 37 shows the gross value added for the environmental goods and services sector in the EU by domain from 2010 to 2020. During the years 2010 to 2020, growth was observed in every major area of the green economy. But much of the growth was attributed to higher GVAs for energy-efficient and renewable energy-related operations, with waste management activities coming in second. The EU-27 economy received a gross value-added contribution from green economy activities in 2020 of EUR 301 billion (2010 values). The goal of EU environment and climate policy is raised by the European Green Deal, which aims to facilitate the shift to a carbon-neutral, circular, green economy by 2050. Consequently, it is anticipated that in the upcoming years, the green economy's share of the EU GDP would rise even more. Applying the concepts of the circular economy to the EU economy, for instance, is predicted to boost GDP by 0.5% by 2030. The "Fit for 55" package, which attempts to boost output from renewable energy sources like solar energy or offshore wind sources and enhance energy efficiency, will also require a sizable rise in economic activity [16].

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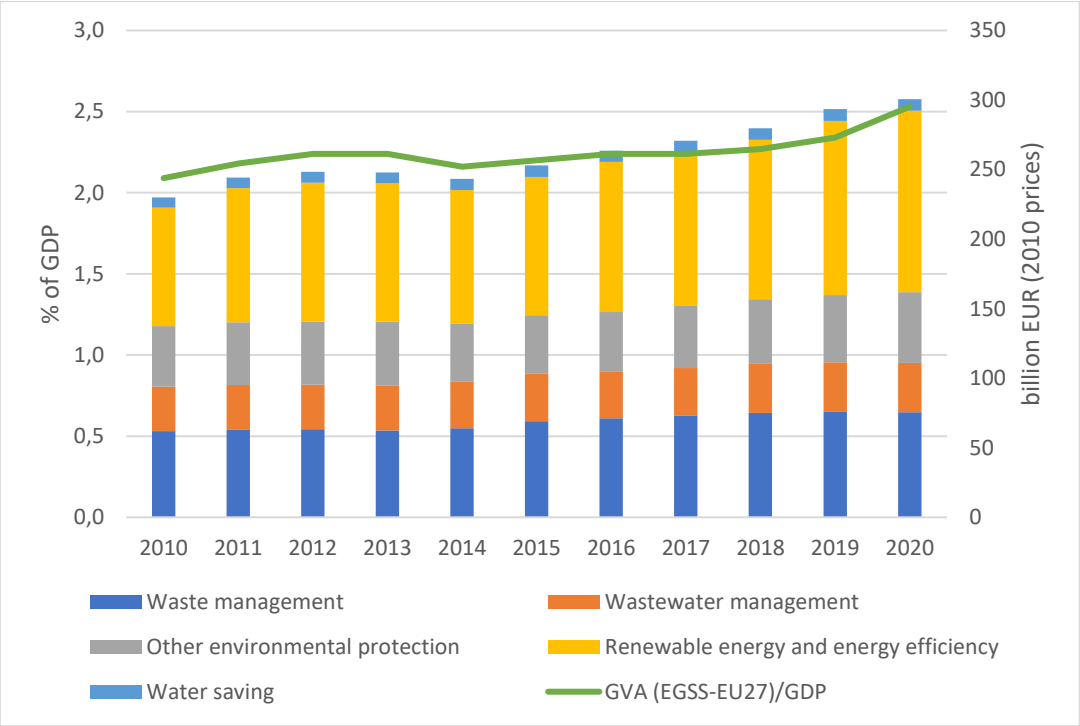

Figure 37 - Gross value added of the EU's environmental goods and services sector by domain, 2010-2020, (source: https://www.eea.europa.eu/data-and-maps/figures/gross-value-added-of-the) [16].

As Bruegel Blueprint illustrates green industrial policy is complex. Defining eight principles for the development of a green industrial policy European Union [15]:

1. Strong governance
2. Tackling geographical fragmentation
3. Managing long-term expectations with a solid EU decarbonisation trajectory
4. Development of sound public-private partnerships
5. Stimulating EU green investment
6. Stimulating EU green science and innovation
7. A global approach
8. Transparent communication

4.7 General information transversal to all

4.7.1 Air pollution

A significant proportion of Europe's urban population lives in cities where EU air quality standards for the protection of human health are regularly exceeded. Air pollution continues to have significant impacts on the health of Europeans, particularly in urban areas. These health impacts have economic costs, cutting short lives, increasing medical costs and reducing productivity through lost working days. The pollutants with the most serious impacts on human health are particulate matter, nitrogen dioxide and ground-level ozone.

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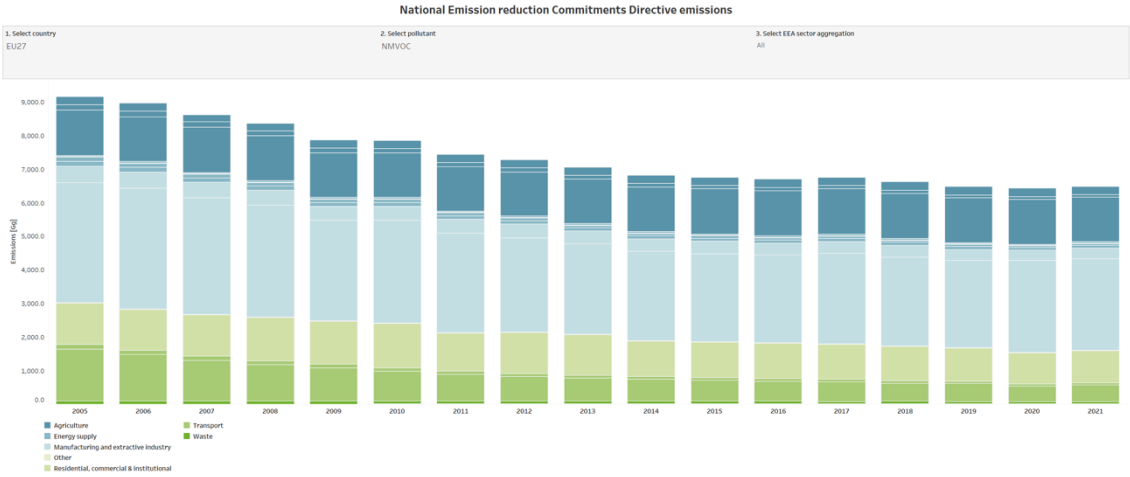

Figure 38 - National Emission reduction Commitments Directive emissions, in euro countries [17].

In figure 39 it can be seen the Air emissions accounts totals bridging to emission inventory totals in the 8 Angie partners countries.

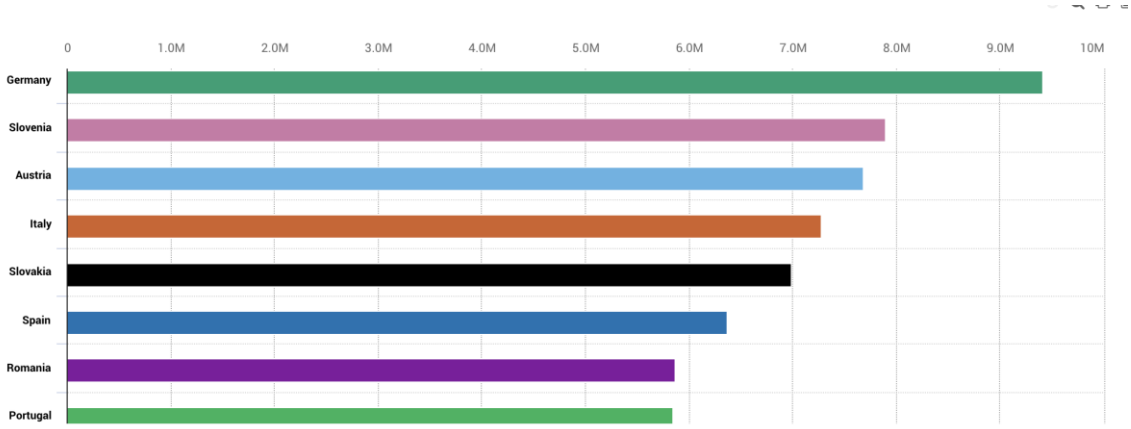

Figure 39 - Air emissions accounts totals bridging to emission inventory totals in the 8 partners countries, in 2022, Unit of measure: Grams per capita [5].

Greenhouse gases emissions into the atmosphere in EU countries, due to economic activities of resident units (businesses, families and government), are represented in figure 40. Those emissions are measured according to the international System of Environmental-Economic Accounting (SEEA).
The greenhouse gases are carbon dioxide (CO2), methane (CH4), nitrous oxide (N2O) and fluorinated gases.

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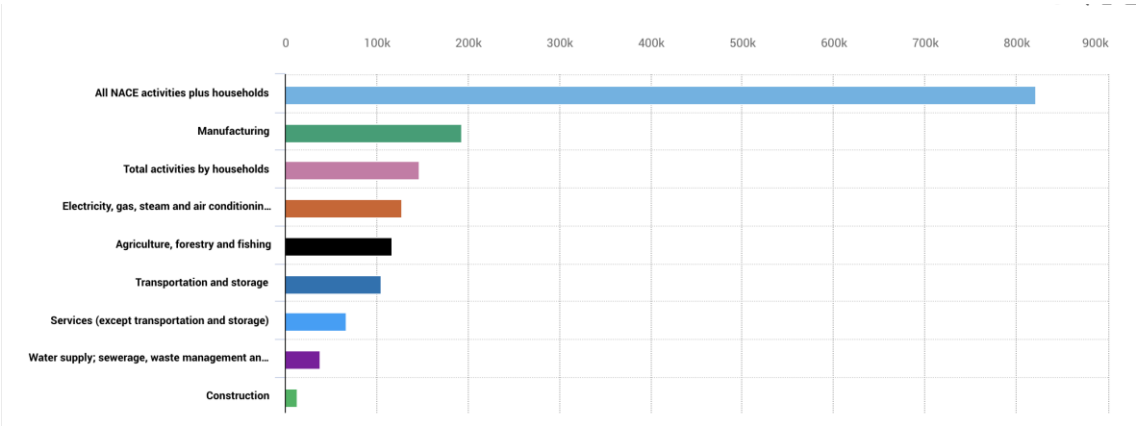

Figure 40 - Air emissions accounts for greenhouse gases by NACE in EU countries, in 2023-Q2. Unit measure Thousand tonnes [5].

4.7.1 Circular material use rate

The circular material use rate (CMR) measures the share of material recovered and fed back into the economy in overall material use. The CMU is defined as the ratio of the circular use of material to the overall material use. The overall material use is measured by summing up the aggregate domestic material consumption (DMC) and the circular use of materials. DMC is defined in economy-wide material flow accounts. The circular use of materials is approximated by the amount of waste recycled in domestic recovery plants minus imported waste destined for recovery plus exported waste destined for recovery abroad. A higher CMU rate value means that more secondary materials substitute for primary raw materials thus reducing the environmental impacts of extracting primary material.
The indicator is part of the EU Sustainable Development Goals (SDG) indicator set. It is used to monitor progress towards SDG on ensuring sustainable consumption and production patterns which is embedded in the European Commission’s Priorities under the 'European Green Deal'.

SDG calls for a comprehensive set of actions from businesses, policy-makers, researchers and consumers to adapt to sustainable practices. It envisions sustainable production and consumption based on advanced technological capacity, resource efficiency and reduced global waste.

The indicator can be considered as part of the global SDG indicator "Proportion of urban solid waste regularly collected and with adequate final discharge out of total urban solid waste generated, by cities".

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Increasing the circularity of Europe's economy is a key priority of the new Industrial Strategy of the EU adopted in 2020 and one of the key elements of the European Green Deal.

The European Commission adopted the new circular economy action plan (CEAP) in March 2020. The EU's transition to a circular economy will reduce pressure on natural resources and will create sustainable growth and jobs. The new action plan announces initiatives along the entire life cycle of products. It targets how products are designed, promotes circular economy processes, encourages sustainable consumption, and aims to ensure that waste is prevented, and the resources used are kept in the EU economy for as long as possible. Building on existing EU policies and legislation, the Circular Economy Package establishes a programme of action with measures covering the whole cycle from production and consumption to waste management. The Package includes commitments on eco-design, waste prevention and reuse, clean material cycles and ambitious quantitative targets on increasing recycling and reducing landfilling, obligations to improve the separate collection of waste, as well as the promotion of efficient use of bio-based resources.

The Sustainable Consumption and Production and Sustainable Industrial Policy (SCP/SIP) Action Plan includes a series of proposals on sustainable consumption and production that will contribute to improving the environmental performance of products and increase the demand for more sustainable goods and production technologies.

This indicator is also included as a headline indicator in the 8th Environment Action Programme and sets a target of doubling the ratio of circular material use by 2030 compared to 2020.

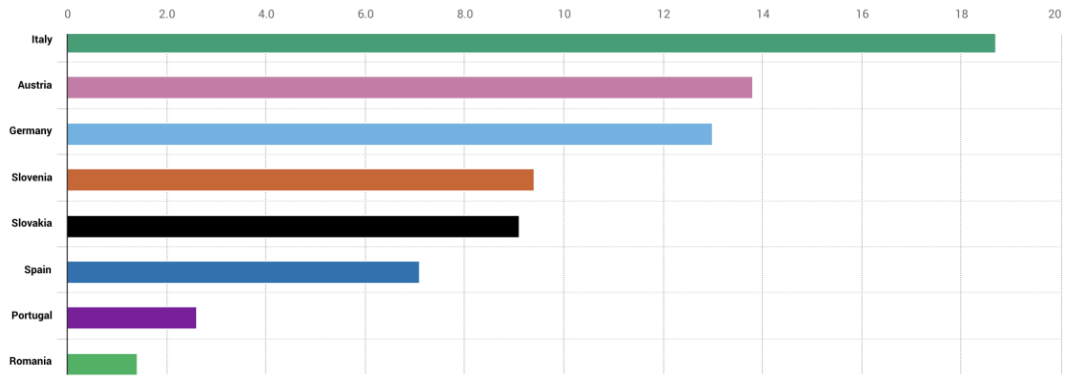

Figure 41 - Circular material use rate in 2022, in the 8 Partners countries, unit of measure percentage [5].

4.7.2 Carbon footprints

´The dataset presents modelling estimates of carbon dioxide (CO2) 'embodied' in products (goods and services) for final demand - also referred to as 'footprints'. The estimates are the result of environmental input-output modelling and cover the entire world economy. The carbon footprint is a measure of how much CO2 was emitted along the full production chain of a product that ends up in the EU as final consumption or

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investment, irrespective of the industry or country where the CO2 emission occurred. These emissions are sometimes referred to as emissions 'embodied' in EU consumption, although they are not literally included in the final products, and these products are not only consumed, but may also be investment goods. Air emission footprints offer a complementary perspective to greenhouse gas inventories and air emissions accounts. The latter two record emissions on the production side, at the origin of the emissions. In contrast, carbon footprints are estimated from the perspective of the final product and where it ends up and are therefore also referred to as consumption-based accounts. The distinction between the three perspectives is further described in Eurostat's page on emissions of greenhouse gases and air pollutants" [1, 5].

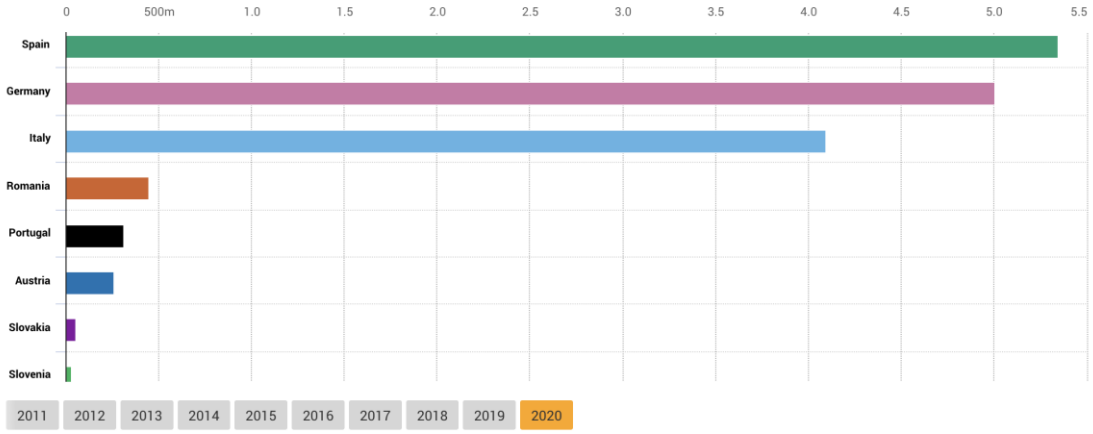

Figure 42 - Carbon dioxide emission footprints. Unit of measure: thousand tonnes.[5].

4.8 Partners Countries contribution to climate exchange

The indicator measures the total amount spent from the annual budget of the EU Member States as well as of the European Commission and the European Investment Bank, in order to contribute to the international 100bn USD commitment for climate finance under the United Nations Framework Convention on Climate Change (UNFCCC). From 2020 onwards the figures refer to commitments made. The 21st UNFCCC Conference of the Parties in Paris urged developed-country Parties again to 'scale up their level of financial support, with a concrete roadmap to achieve the goal of jointly providing USD 100 billion annually by 2020 for mitigation and adaptation and to further provide appropriate technology and capacity-building support.

The indicator is part of the EU Sustainable Development Goals (SDG) indicator set. It is used to monitor progress towards SDG 13 on climate action; which is embedded in the European Commission's Priorities under the European Green Deal. SDG 13 aims to strengthen countries' resilience and adaptive capacity to climate-related hazards and

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natural disasters by integrating climate change mitigation and adaptation measures into national strategies, policies and planning.

Indicator can be considered as identical to global SDG indicator 'Amounts provided and mobilized in United States dollars per year in relation to the continued existing collective mobilization goal of the $100 billion commitment through to 2025'

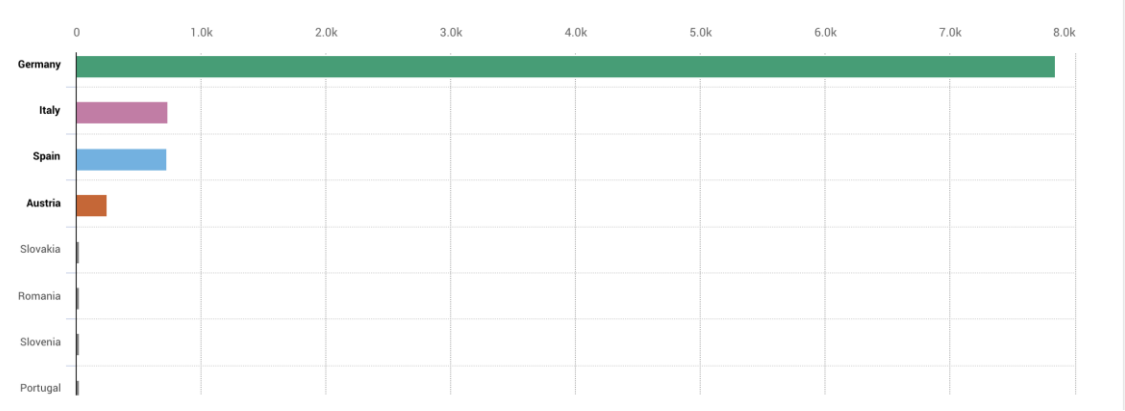

Figure 43 - Contribution to the international 100bn USD commitment on climate related expending in 2021, in the 8 partners courtiers. Unit of measure: Million euro [5].

Converting from the conventional linear economic model to a circular one would not only save the EU's economy, but also greatly minimize the unfavourable impact on the environment. As a result, the circular economy (CE) has gained prominence as one of the most powerful and contemporary pushes toward sustainability. An economy may become circular by adopting circularity into its business models. which is dependent on both authorities and their actions as well as commercial entities.

The EU uses and promotes the idea of a circular economy to boost competition shield firms from inadequate resources and support the development of new possibilities and emerging economies. Converting from the conventional linear economic model to a circular one would not only save the EU's economy, but also greatly minimize the unfavourable impact on the environment. As a result, the circular economy (CE) has gained prominence as one of the most powerful and contemporary pushes toward sustainability. An economy may become circular by adopting circularity into its business models. which is dependent on both authorities and their actions as well as commercial entities. The EU uses and promotes the idea of a circular economy to boost competition shield firms from inadequate resources and support the development of new possibilities and emerging economies.

With the emergence and implementation of the European Green Deal, a series of measures came into force aimed at propelling the circular economy model, restoring the proper functioning of ecosystems, reducing pollution, and offering added value to products that would have been lost in the supply chain. However, adapting to the new rules is not at all within reach as a series of measures and sacrifices are required, thus adapting to the new laws is not at all feasible.

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The circular economy in Romania is in the development phase, but there are initiatives at both the government and private levels that can support and contribute to its development. The promotion of ecological products. The selective collection and recycling of waste are some of the programs implemented, but there are many challenges to face, to reach a high level of implementation of the circular economy in Romania.

Among the benefits that the circular economy can bring are the reduced costs related to waste management the reuse of resources and the increase in productivity of their use. Agriculture represents an important ROLE in the process of implementing the circular economy through the implementation of sustainable agricultural practices, which aim to reduce energy and water use, protect biodiversity, and contribute to the improvement of soil quality. The circular economy can use recyclable technologies and materials instead of non-biodegradable ones but also renewable energy sources for agricultural activity [9].

5 Conclusions

This activity was carried out in the purpose of understanding, analysing and identifying the most relevant needs, that will help the consortium to define topics to be used within the curriculum, which is part of the deliverable D2.1.2.
In summary the most relevant topics taken from this analysis, by domain, are:

Construction:
Energy Efficiency of the buildings
Sustainable Materials
Sustainable urban planning
thermal performance of the buildings
Climate Neutral New Buildings
Waste from construction

Industrial activities:
Greenhouse gases Emissions
Resource Efficiency/circular economy
Environmental Performance Index (EPI) and Environmental Compliance
Energy efficiency in industry
waste from manufacturing
Raw material consumption (RMC)
Green Industrial policy
Air emissions in industry
Circular material use

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Transports:

Emissions of fuels combustion/greenhouse gas emissions

Inadequate transportation infrastructure

Alternative Fuels/sustainable transportation

Batteries

advanced biofuels

Modal shift in passenger and freight transport to environmentally friendly transport modes

Increase fuel efficiency in road transport

Increase the share of clean energy sources in road transport

alternative fuels infrastructure regulation

energy consumption in transport by type of fuel

Policies and measures by transport sector

Waste:

Improper waste disposal and landfill use

Plastic Pollution

Circular Economy including:

- Packaging and plastics
- Sustainable textiles
- Electronics and ICT
- Construction and buildings
- Batteries and vehicles
- The food chain.
- Critical raw materials
- Repairing and reusing goods

Source of waste generation

Waste treatment/Waste Technologies:

- Recovery -Recycling and backfilling (excluding energy recovery)
- Energy recovery
- Recovery - backfilling
- Recovery - recycling
- Incineration (without energy recovery)
- Disposal - landfill
- Disposal - other
- Landfill / disposal

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Energy production:

Renewable Energy
Energy Storage
Grid Modernization
Energy efficiency,
Energy security,
Energy production:

- crude oil,
- petroleum products,
- natural gas and manufactures gases,
- electricity and derived heat,
- solid fossil fuels,
- renewables

Residential activities:

Energy Consumption (heat pumps, appliances and lightening systems,)
Waste production/Reduction
Water waste/Conservation
reduction in the consumption
waste from households
waste from textiles
Municipal waste
Responsible consumption

Agriculture:

Pesticide Use
Water consumption/withdrawals in agriculture
Agriculture Sustainability
Gas emissions from agriculture
Soil loss by water erosion
Energy/resources consumption

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